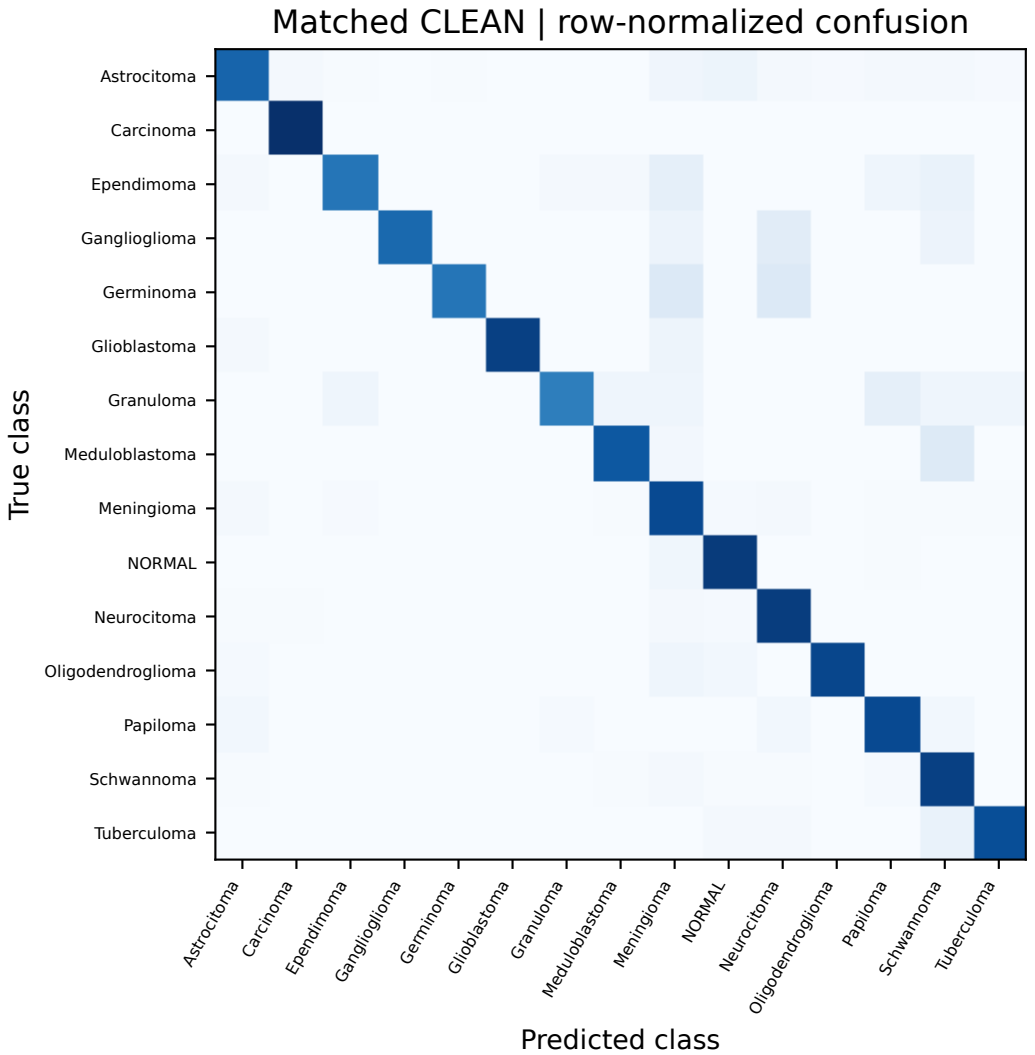
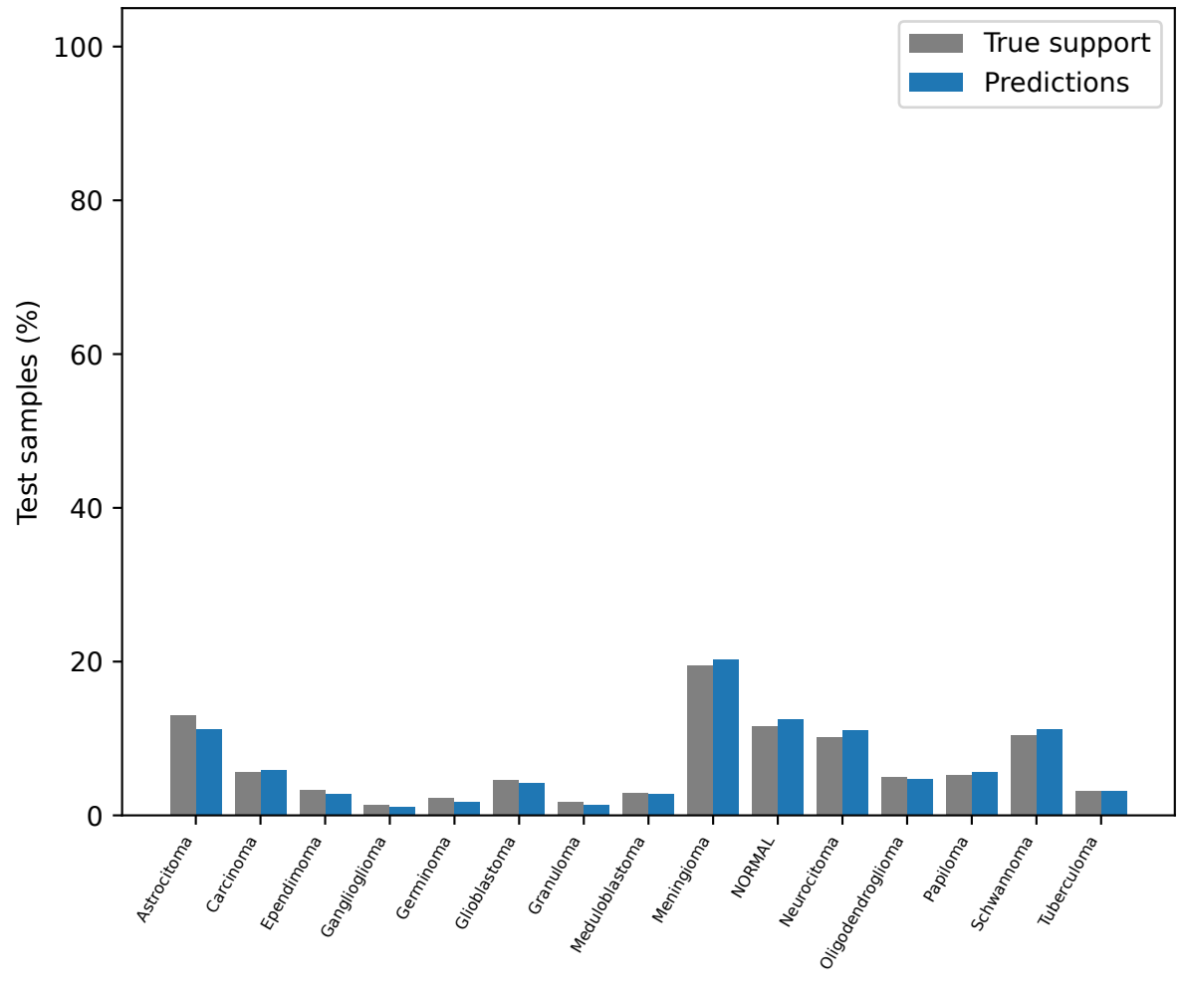
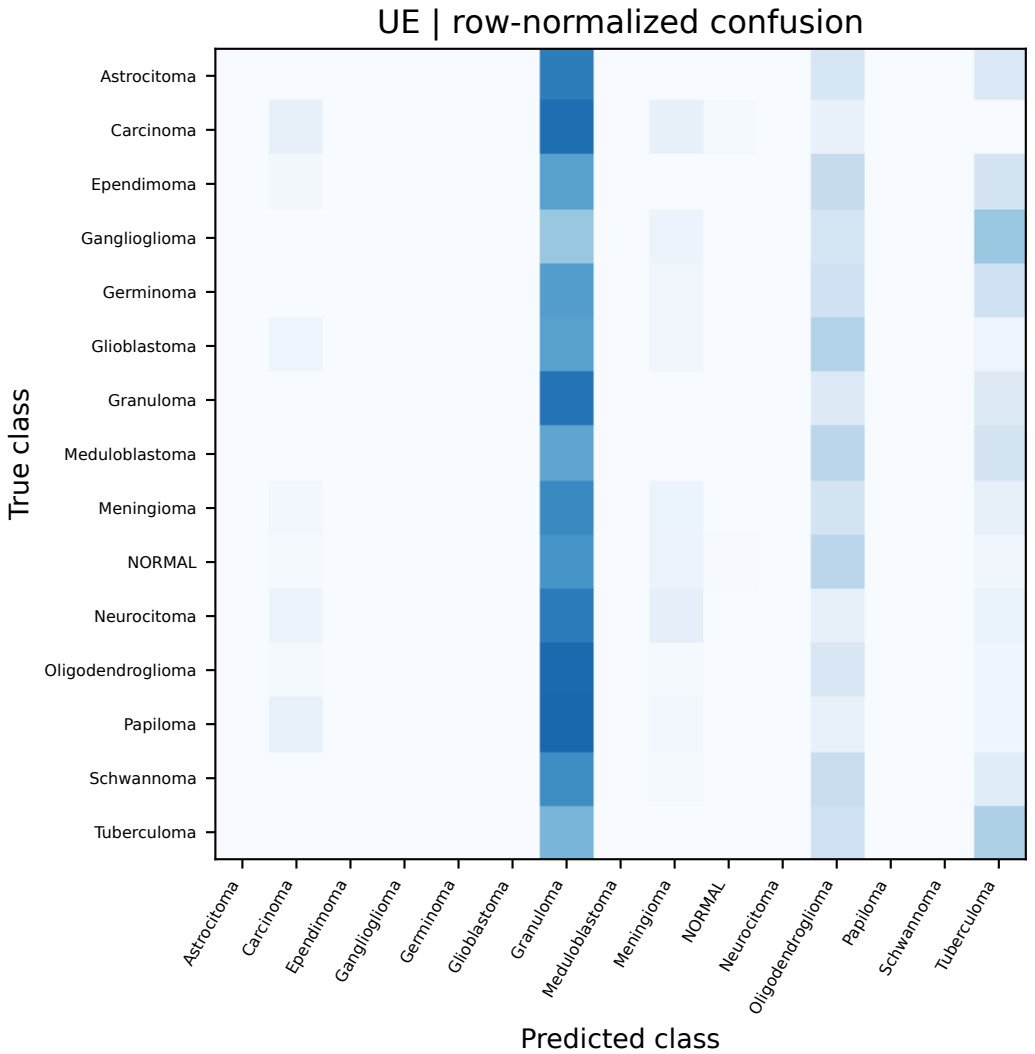
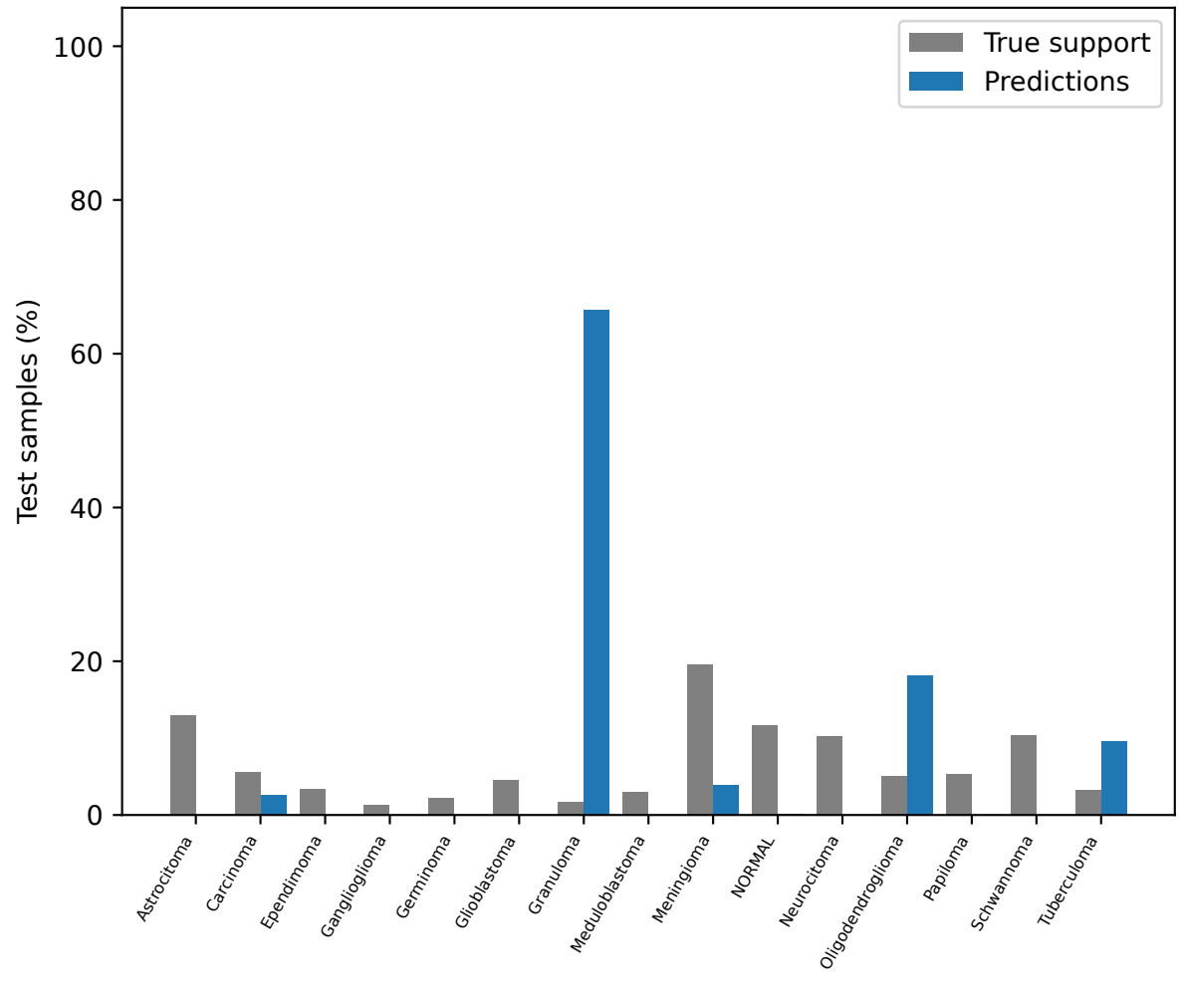
MRI15 | EM

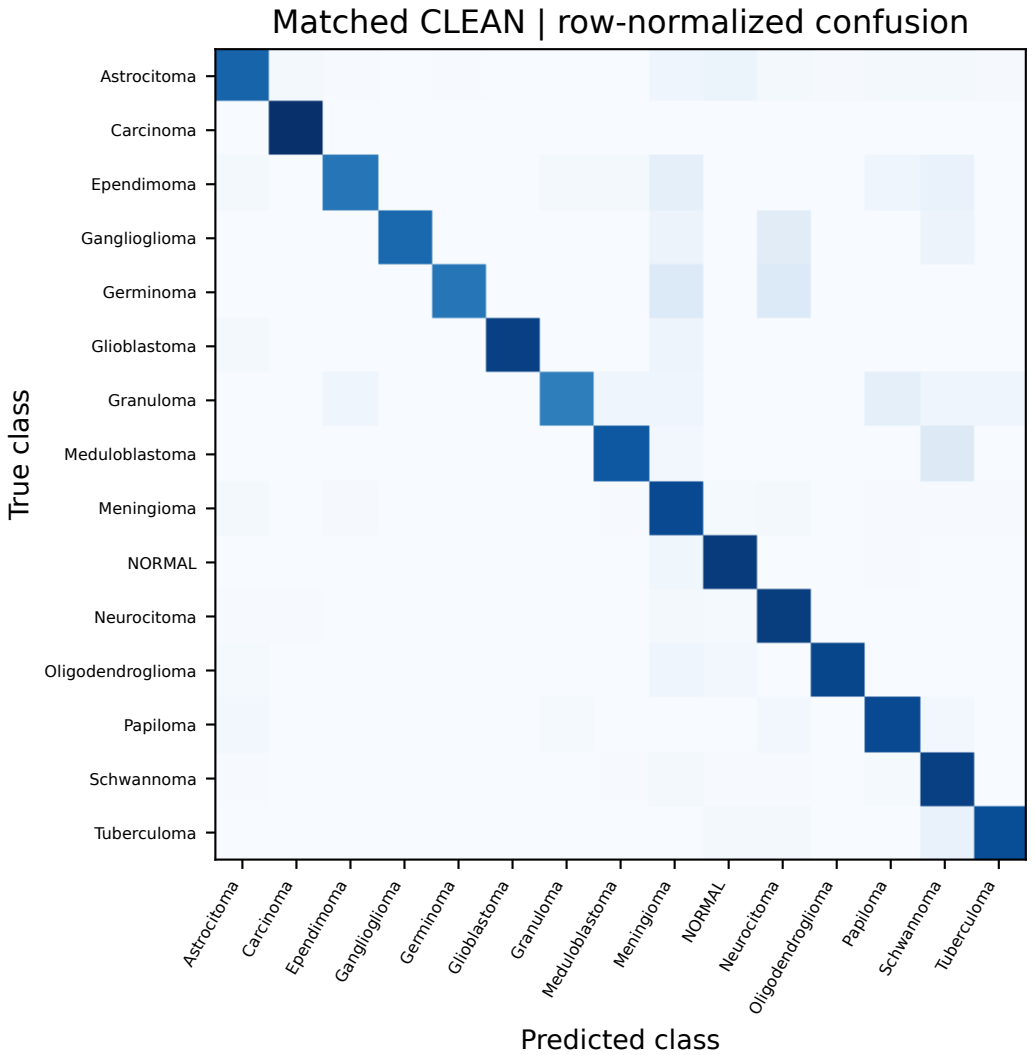

Matched CLEAN | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

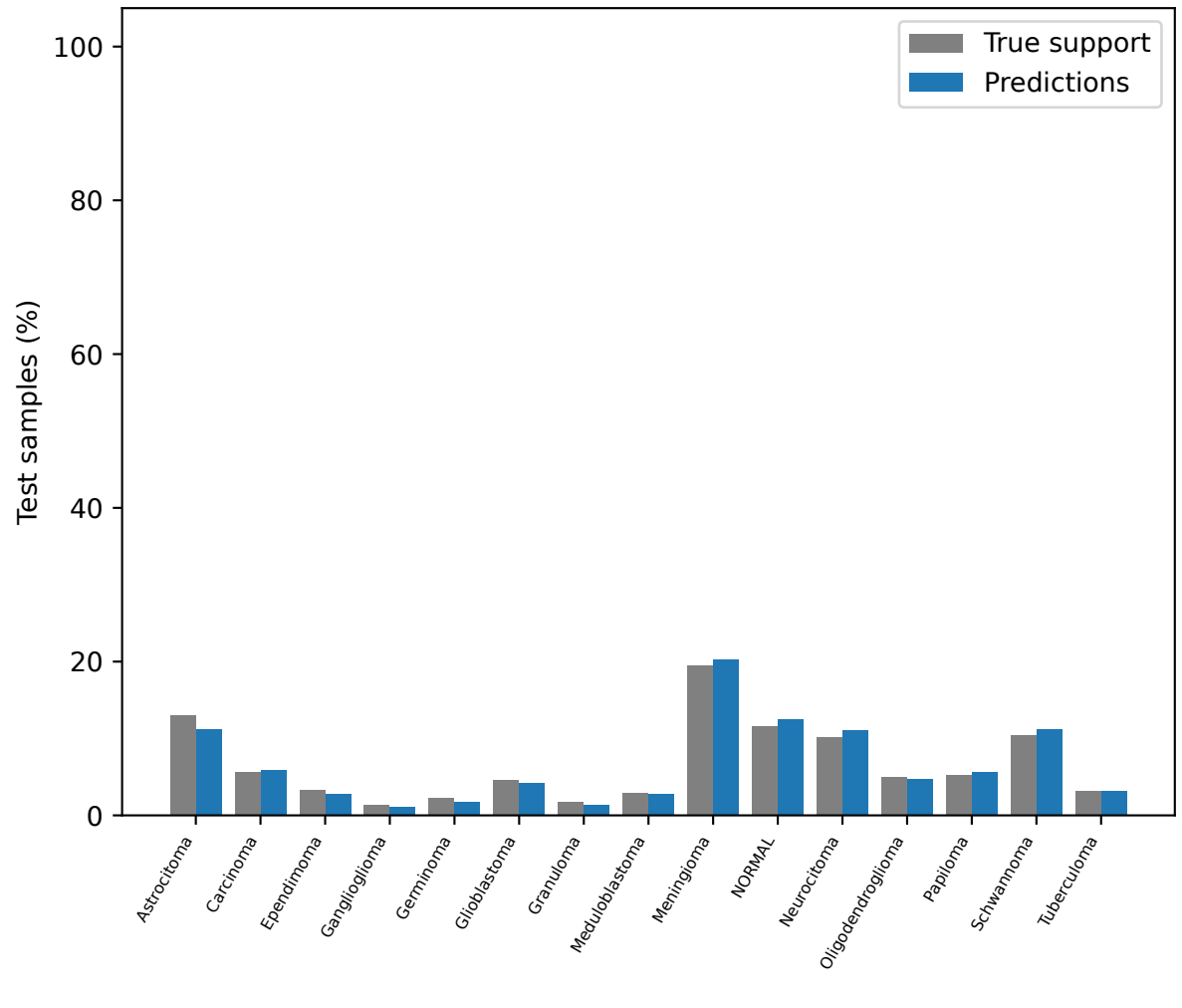

Test samples (%)
True support
Predictions
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

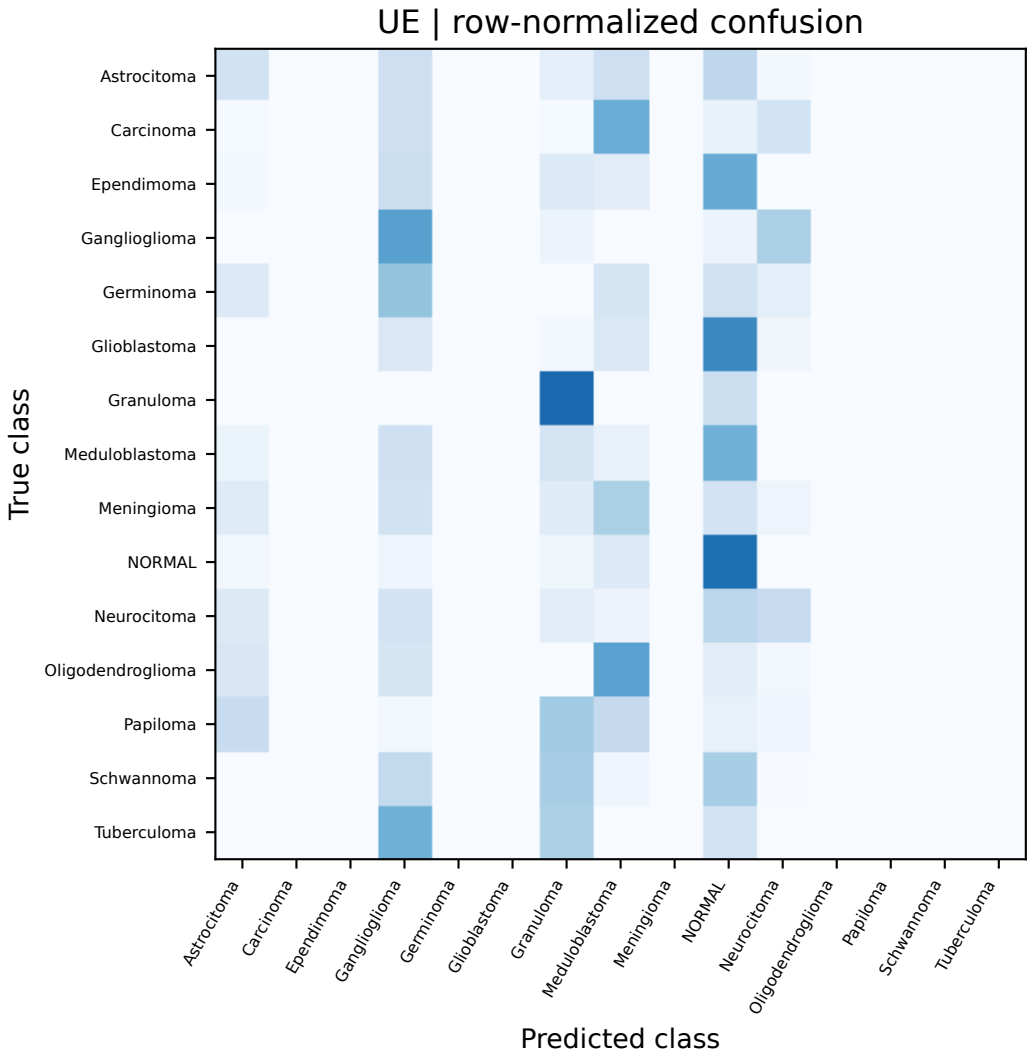

UE | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

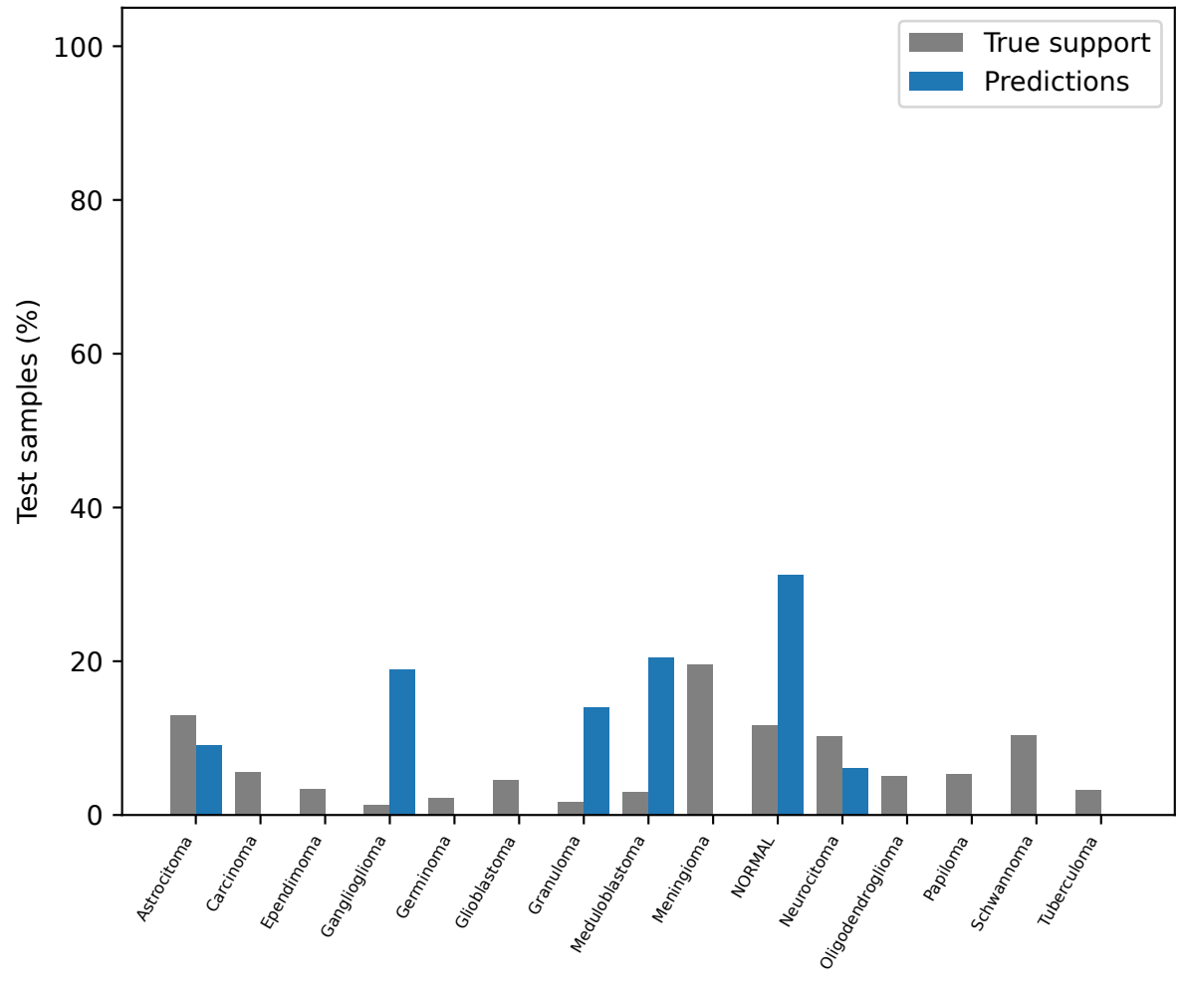

Test samples (%)
True support
Predictions
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

MRI15 | REM

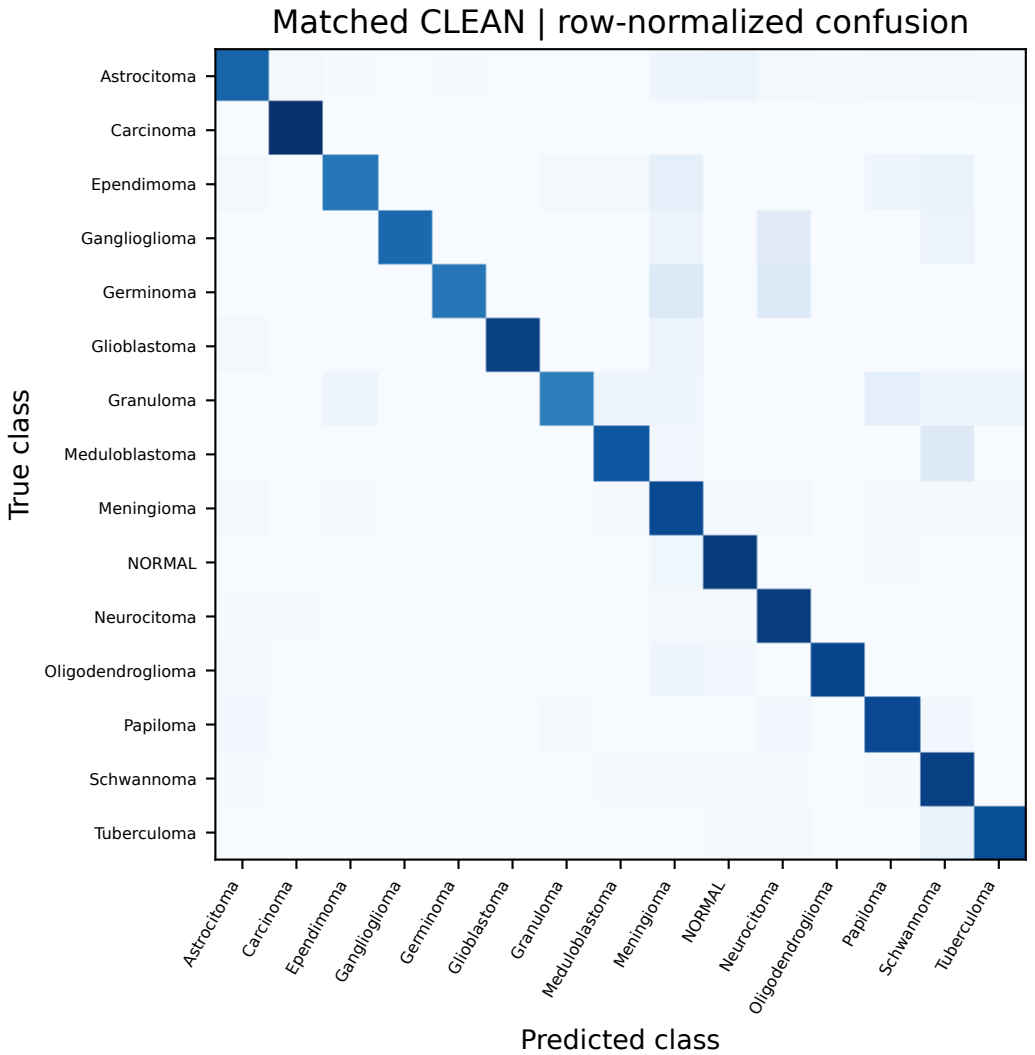

Matched CLEAN | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

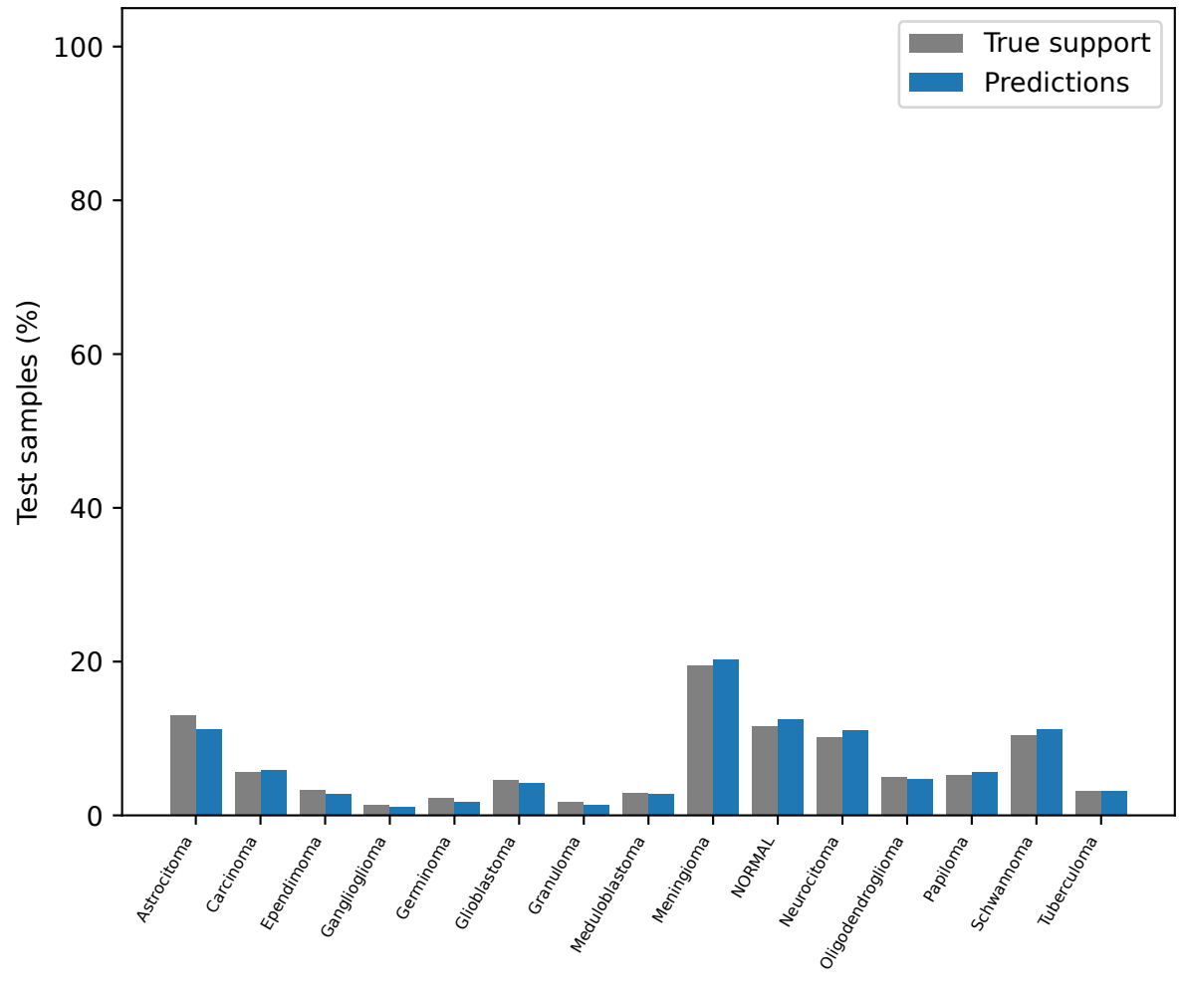

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

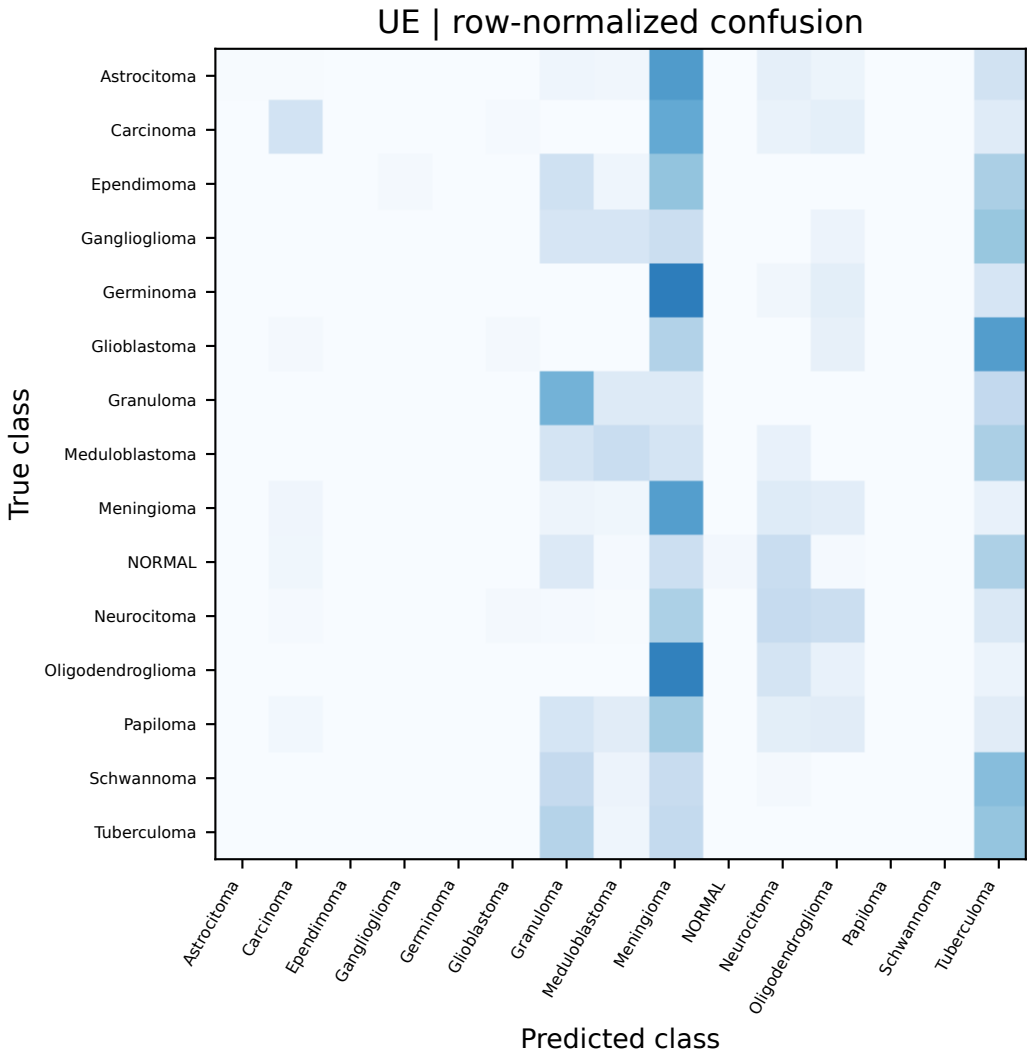

UE | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

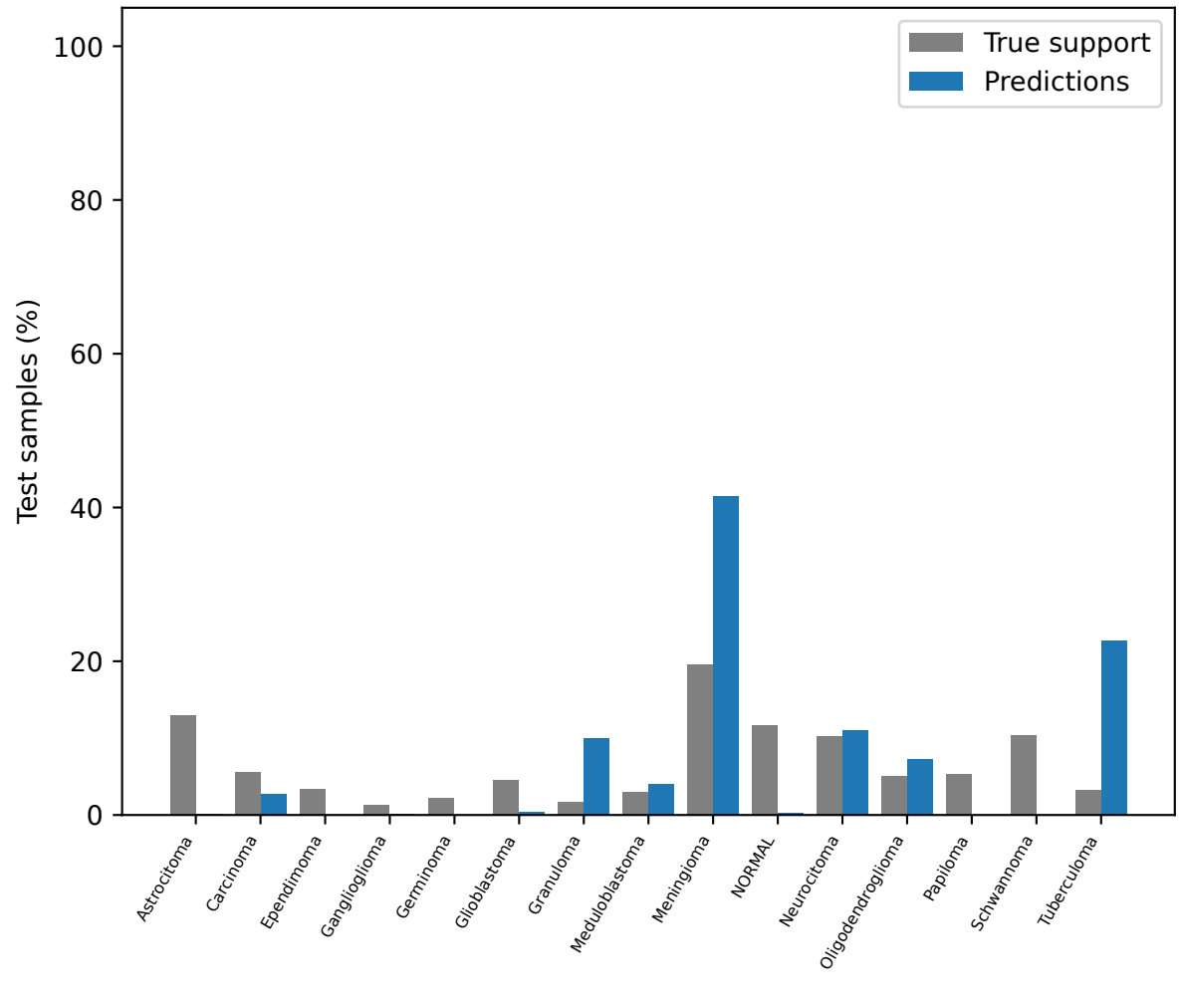

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

MRI15 | SEM

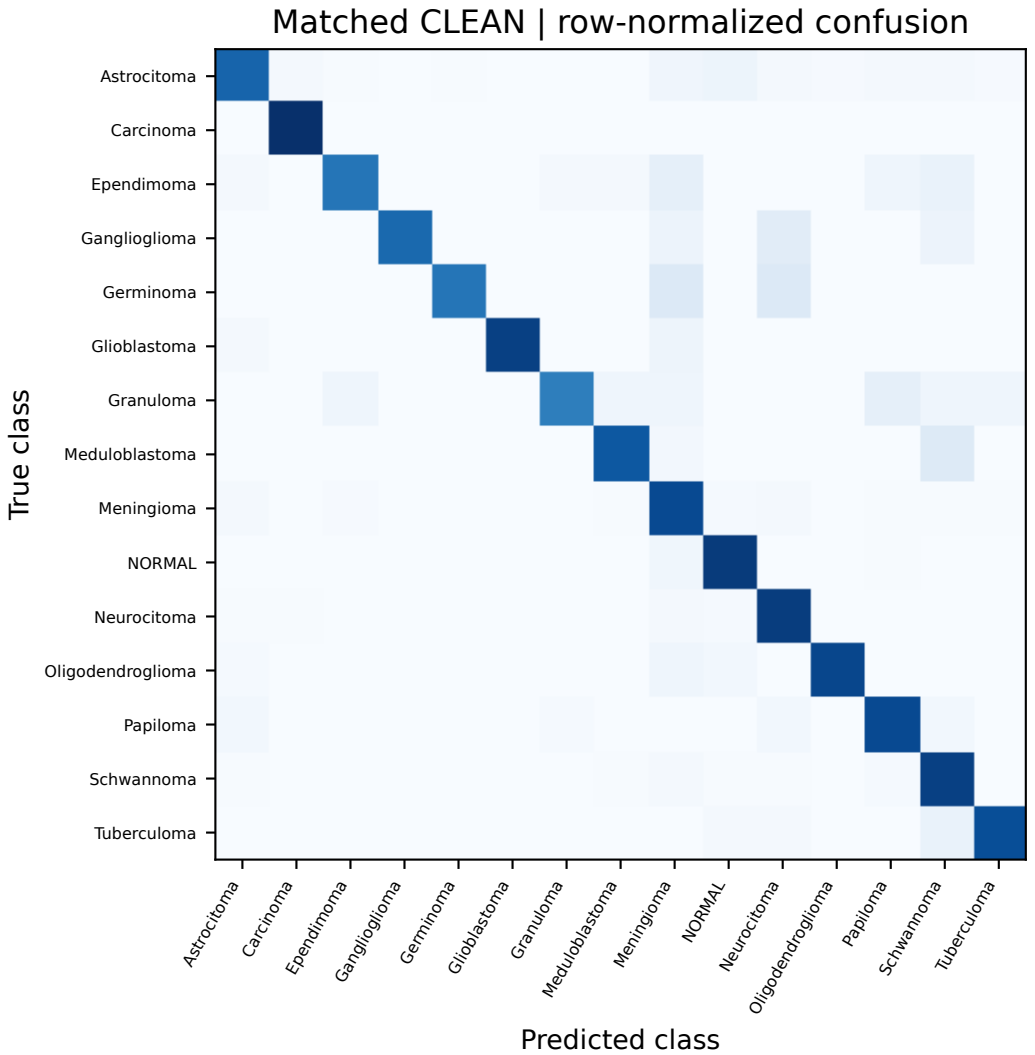

Matched CLEAN | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

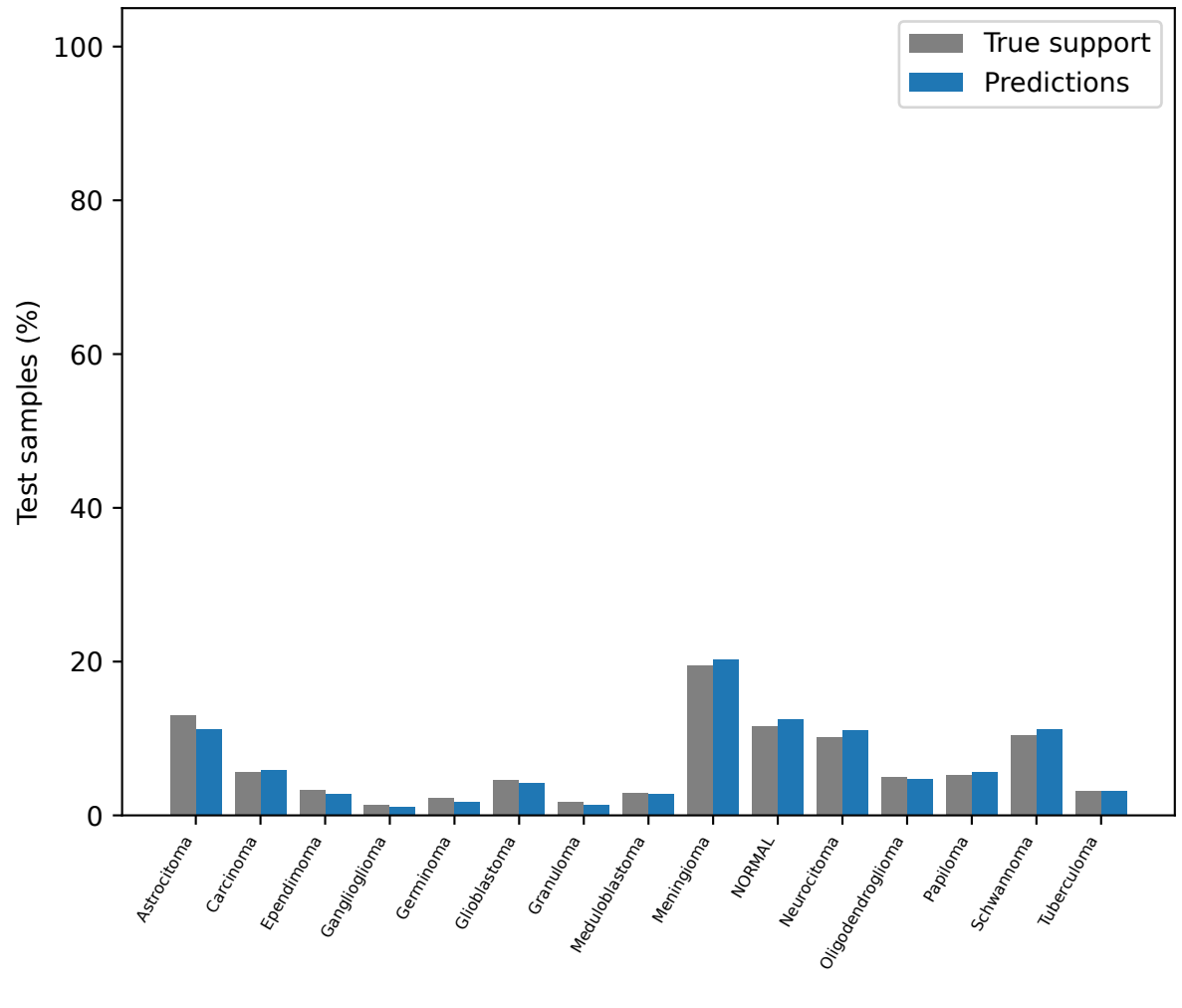

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

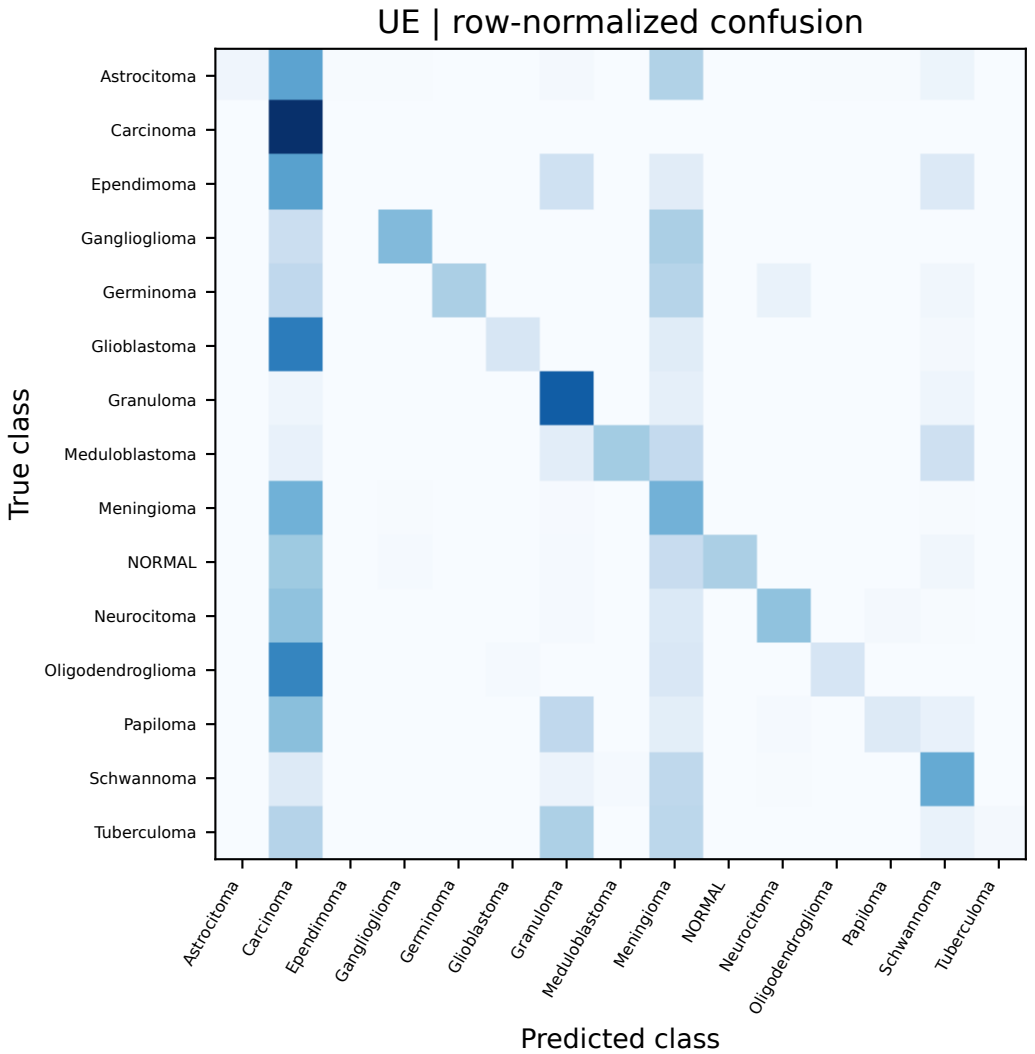

UE | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

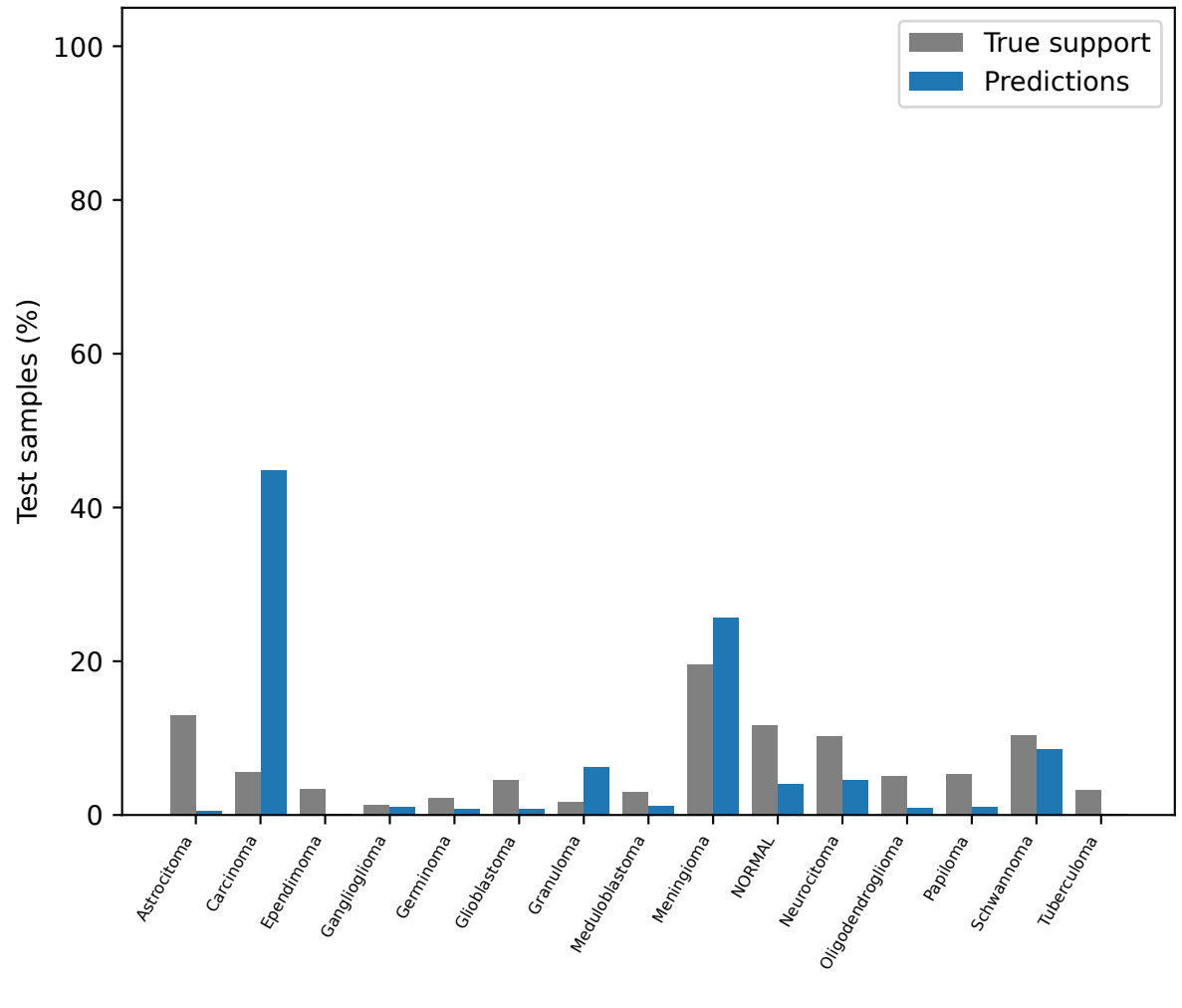

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

MRI15 | GUE

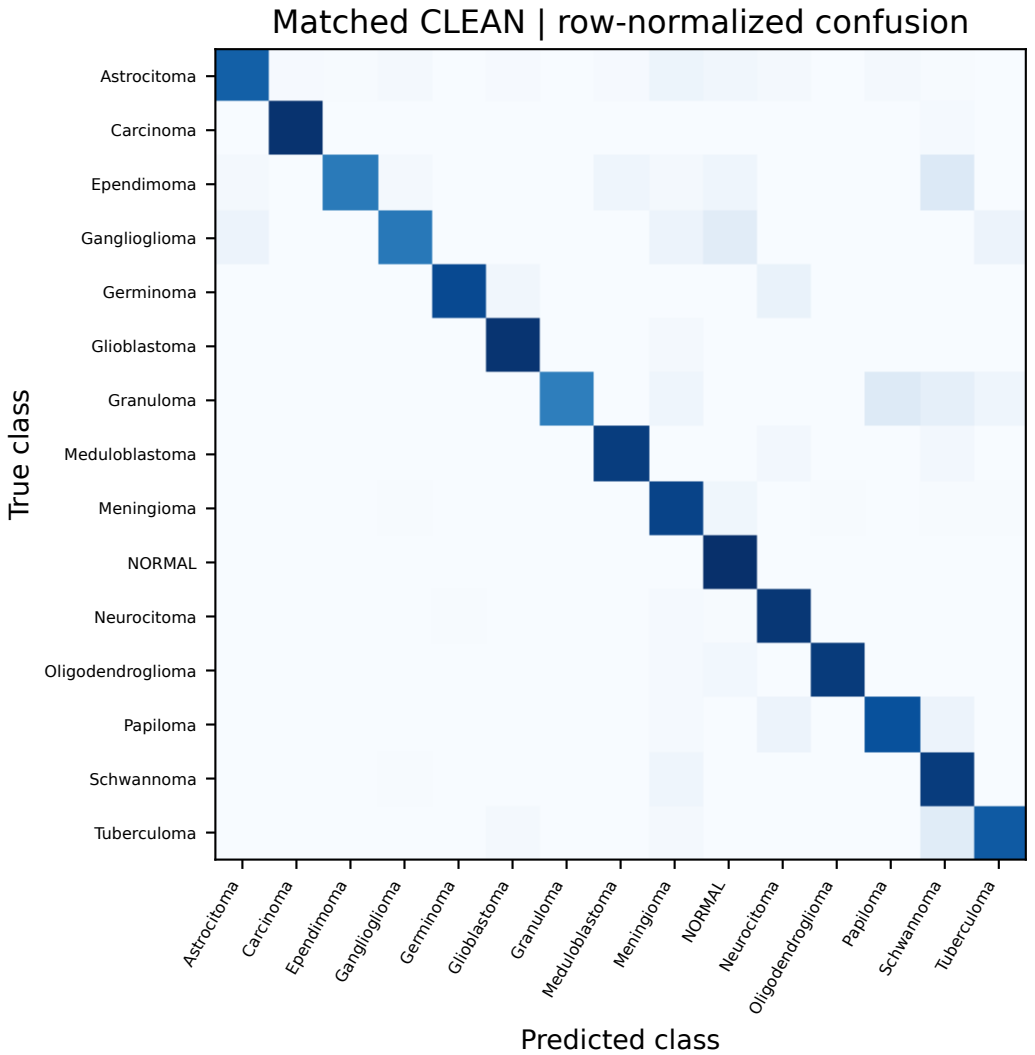

Matched CLEAN | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

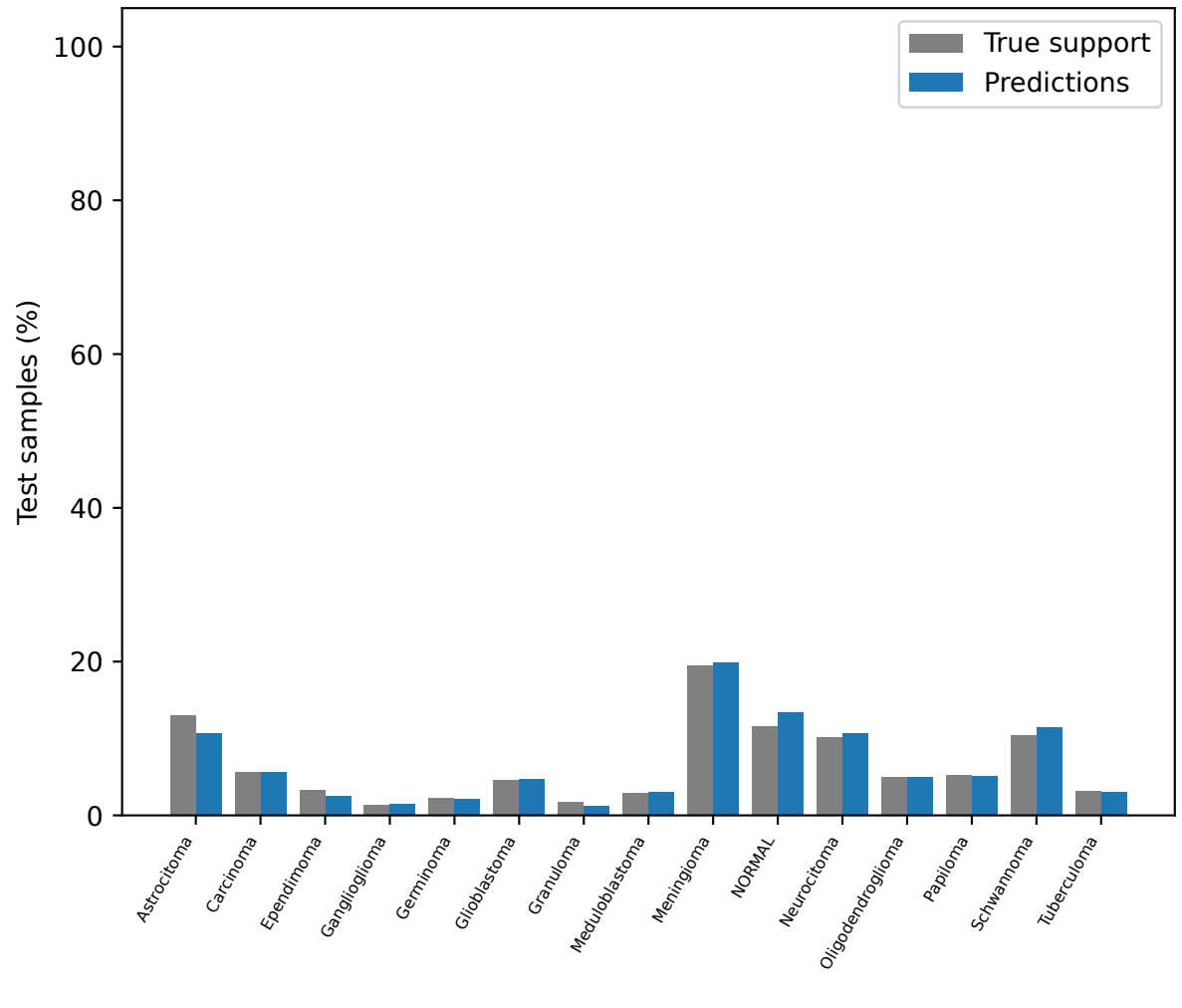

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

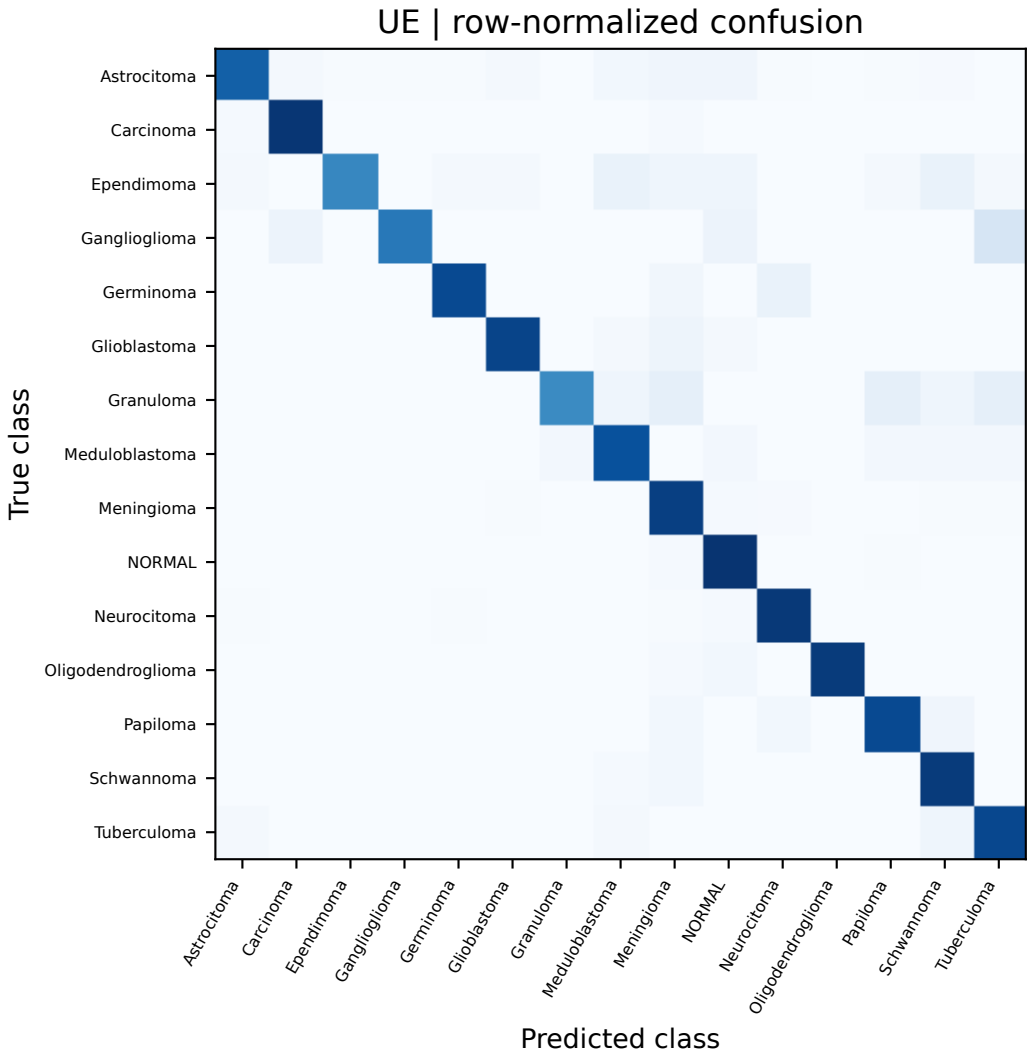

UE | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

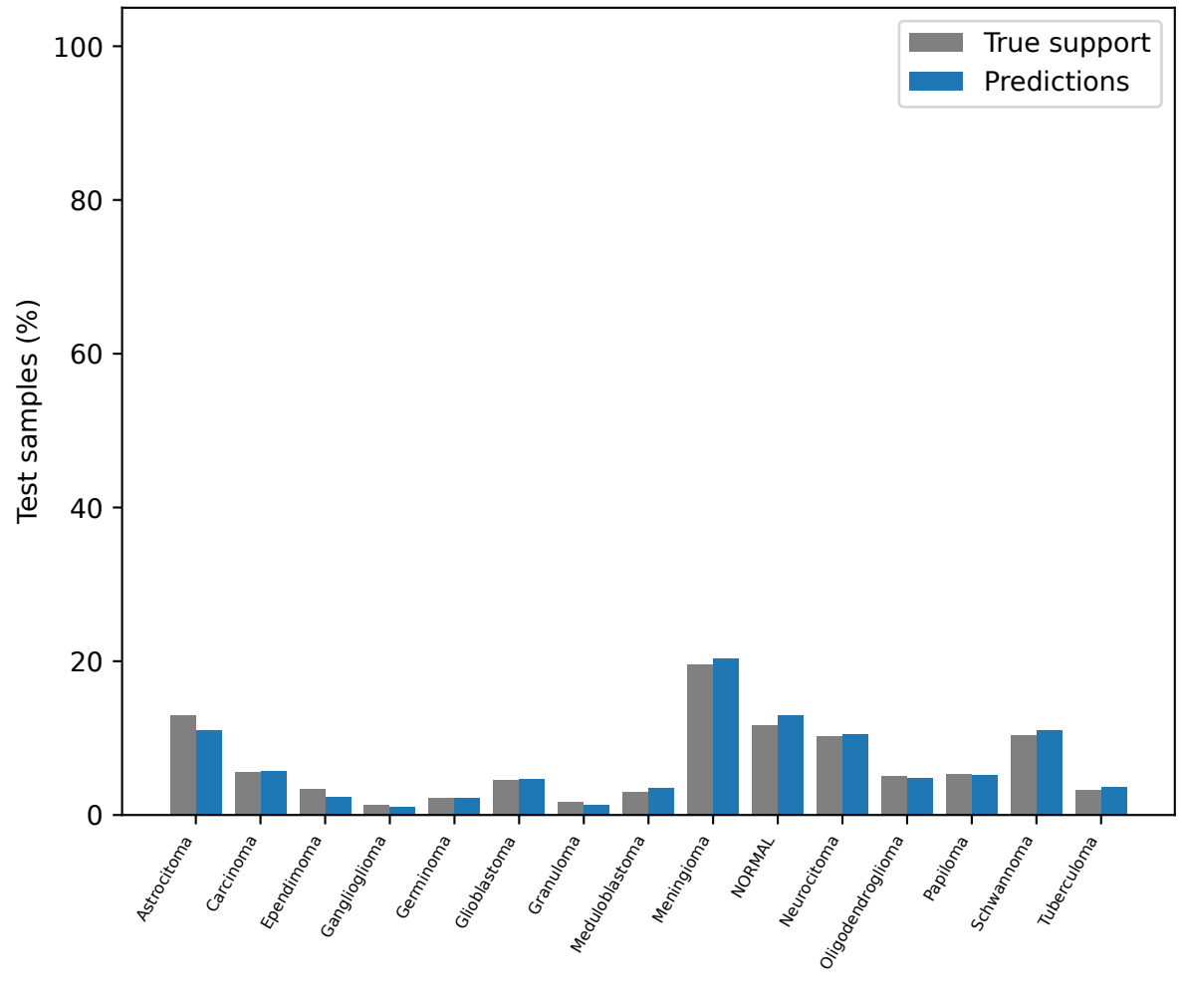

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

MRI15 | AR

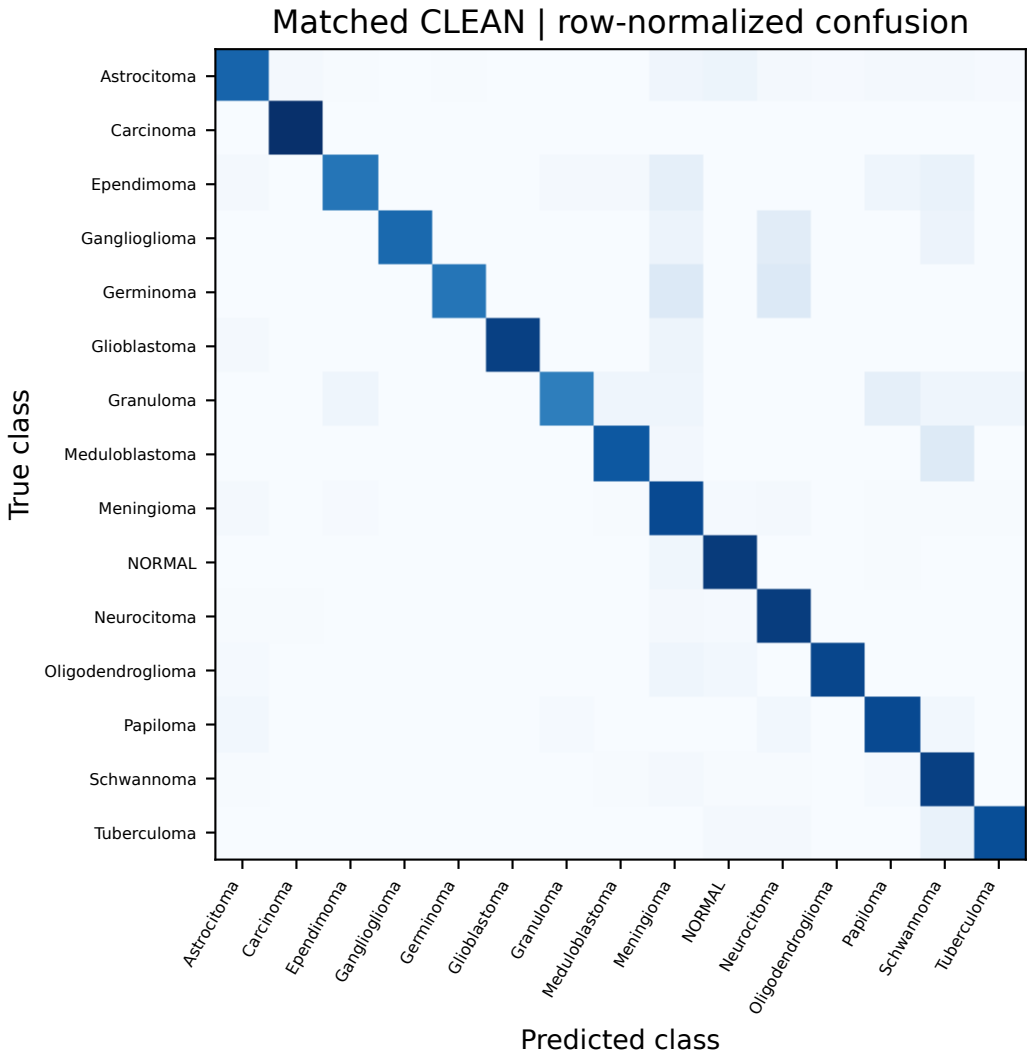

Matched CLEAN | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

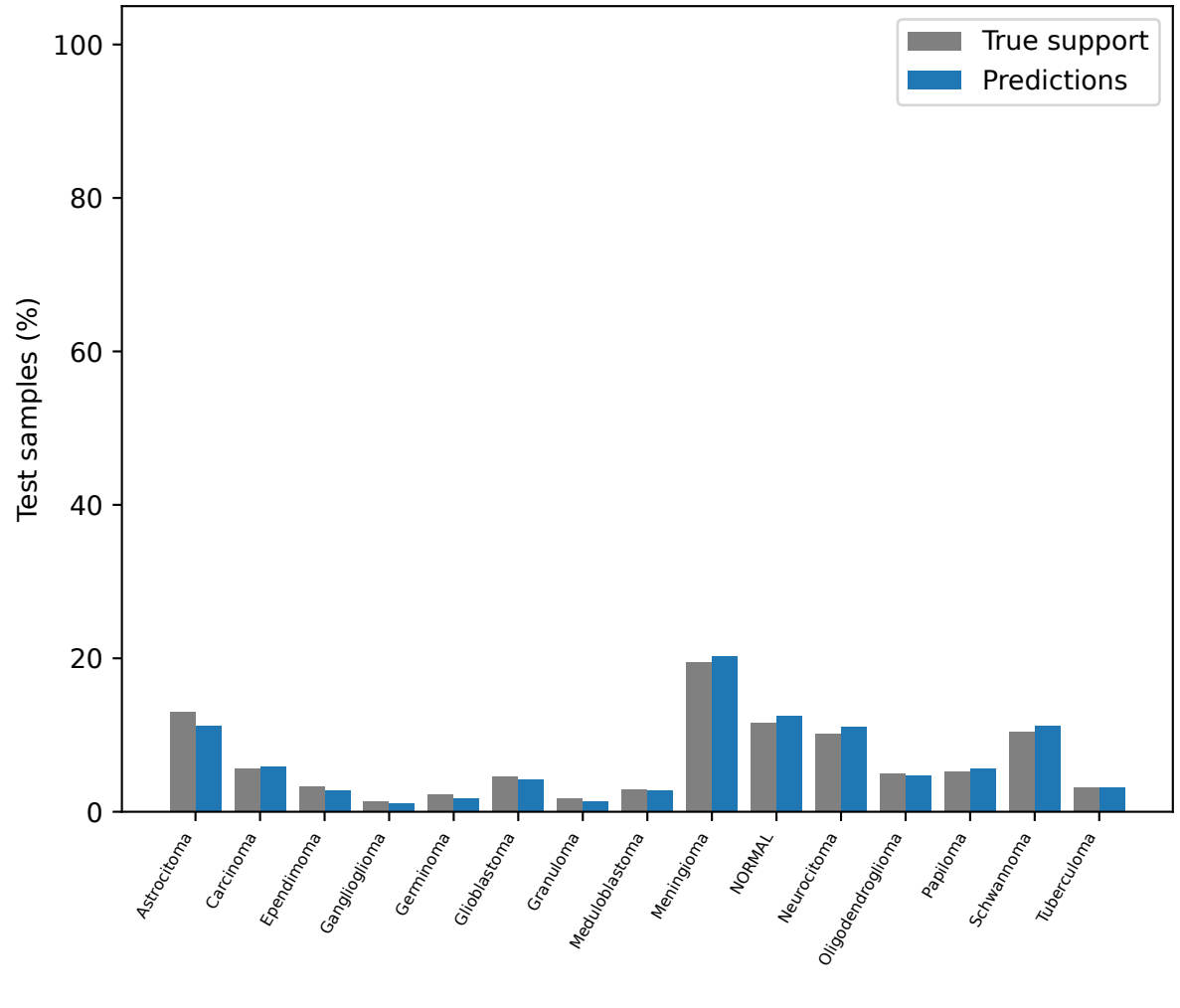

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

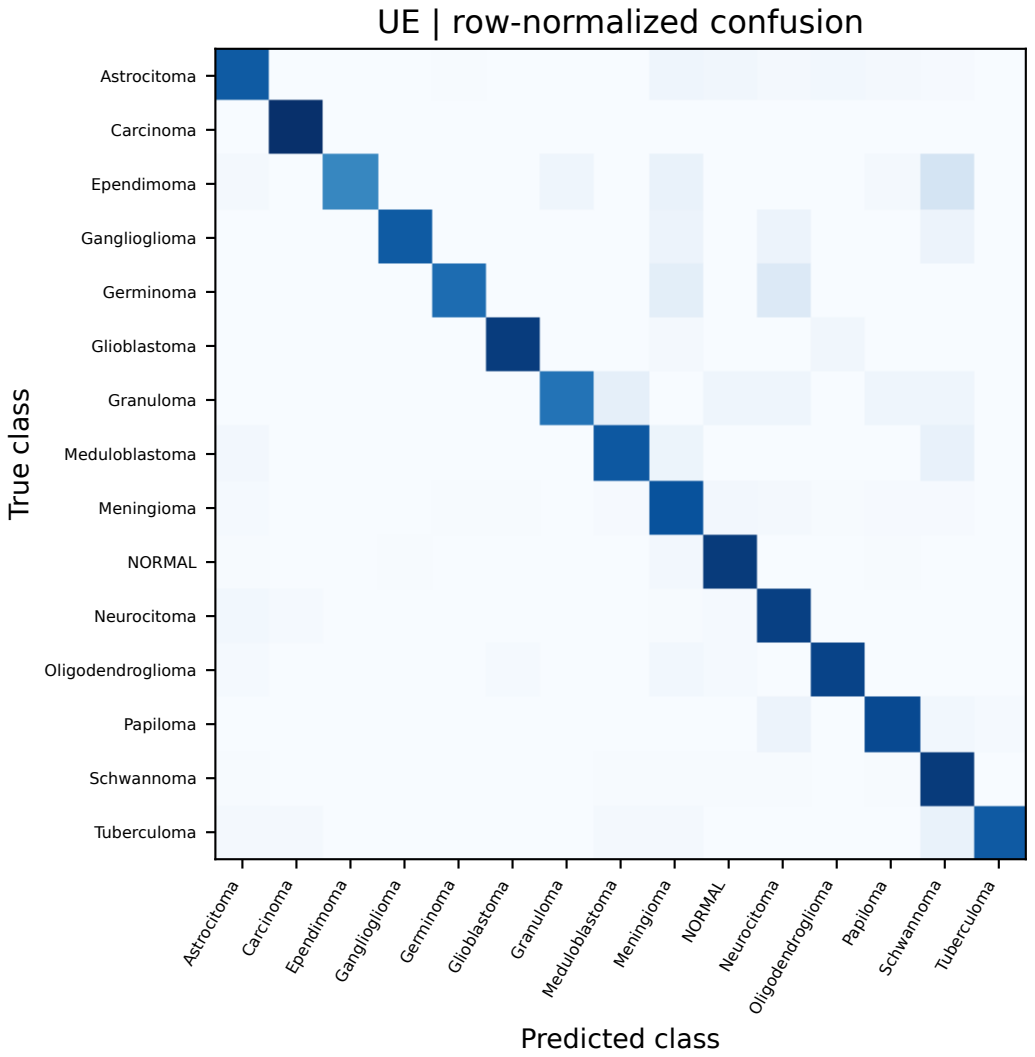

UE | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

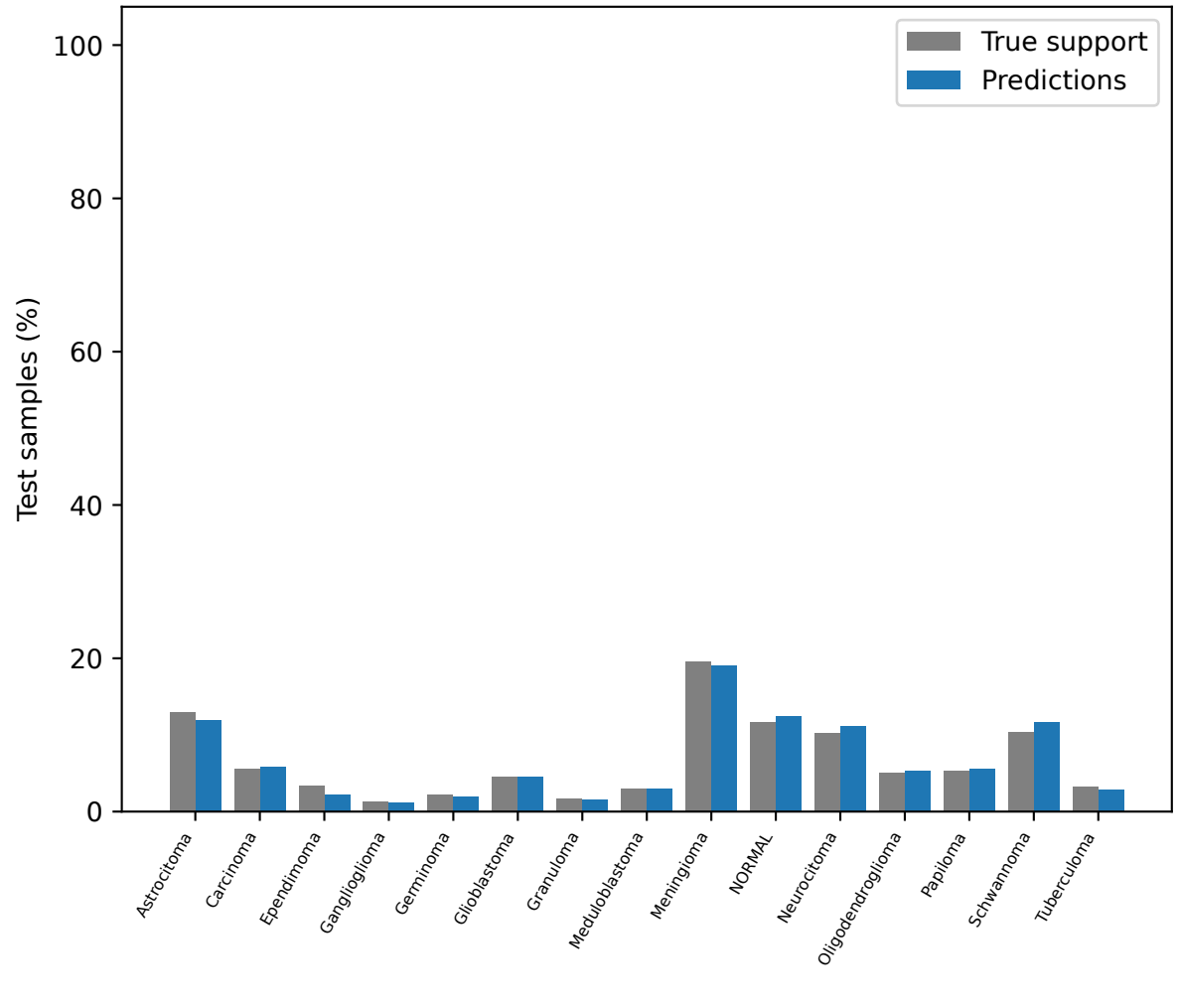

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

MRI15 | LSPNormal

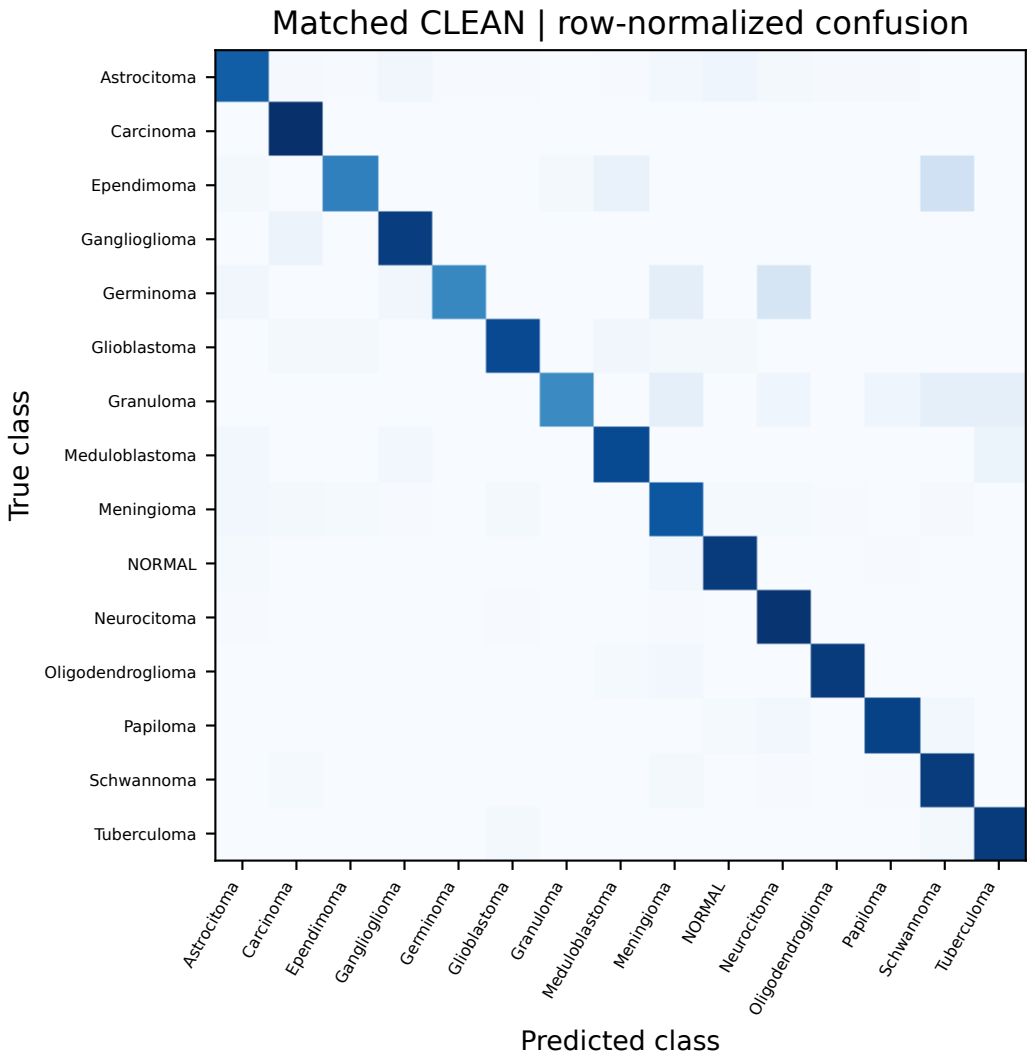


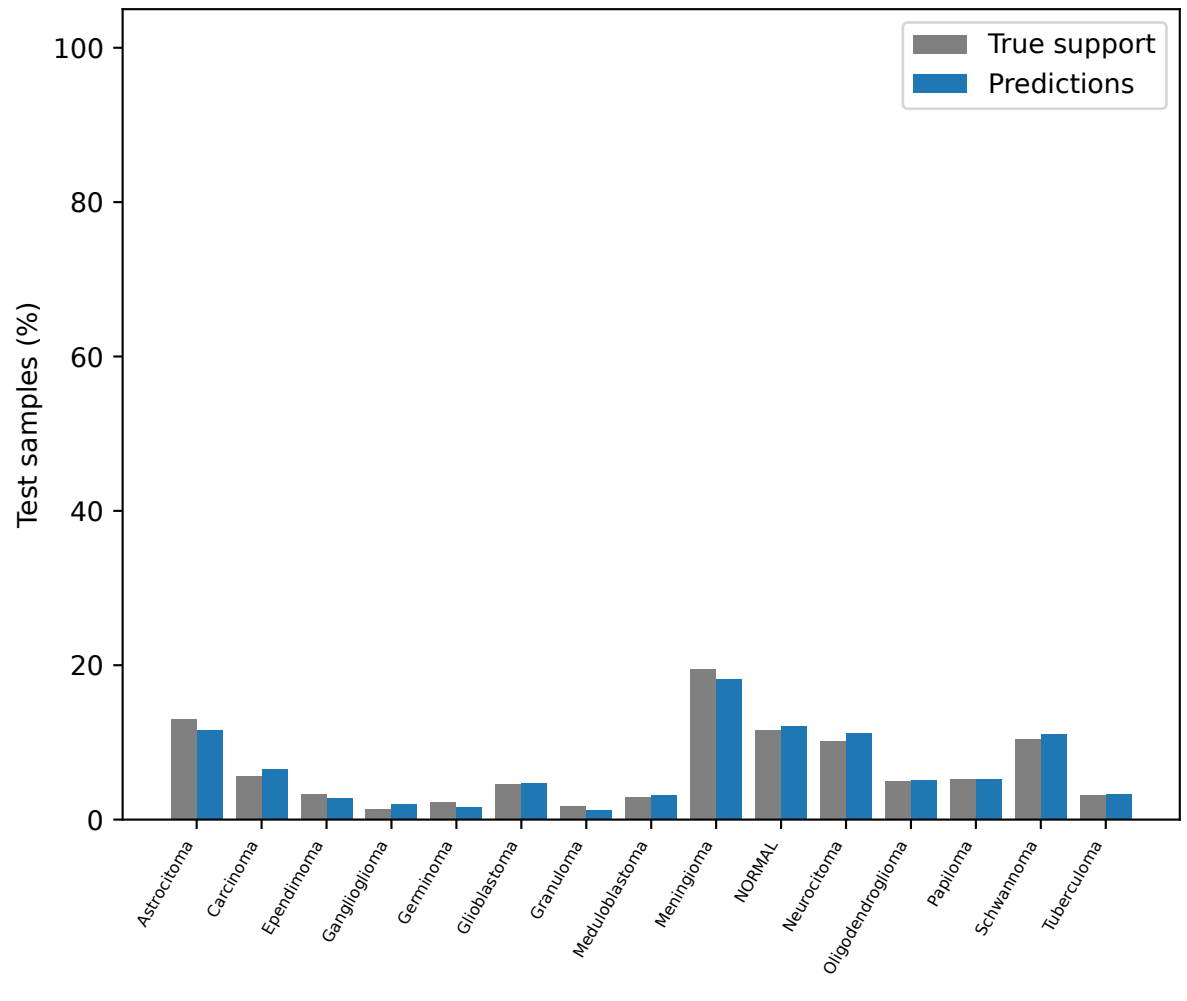


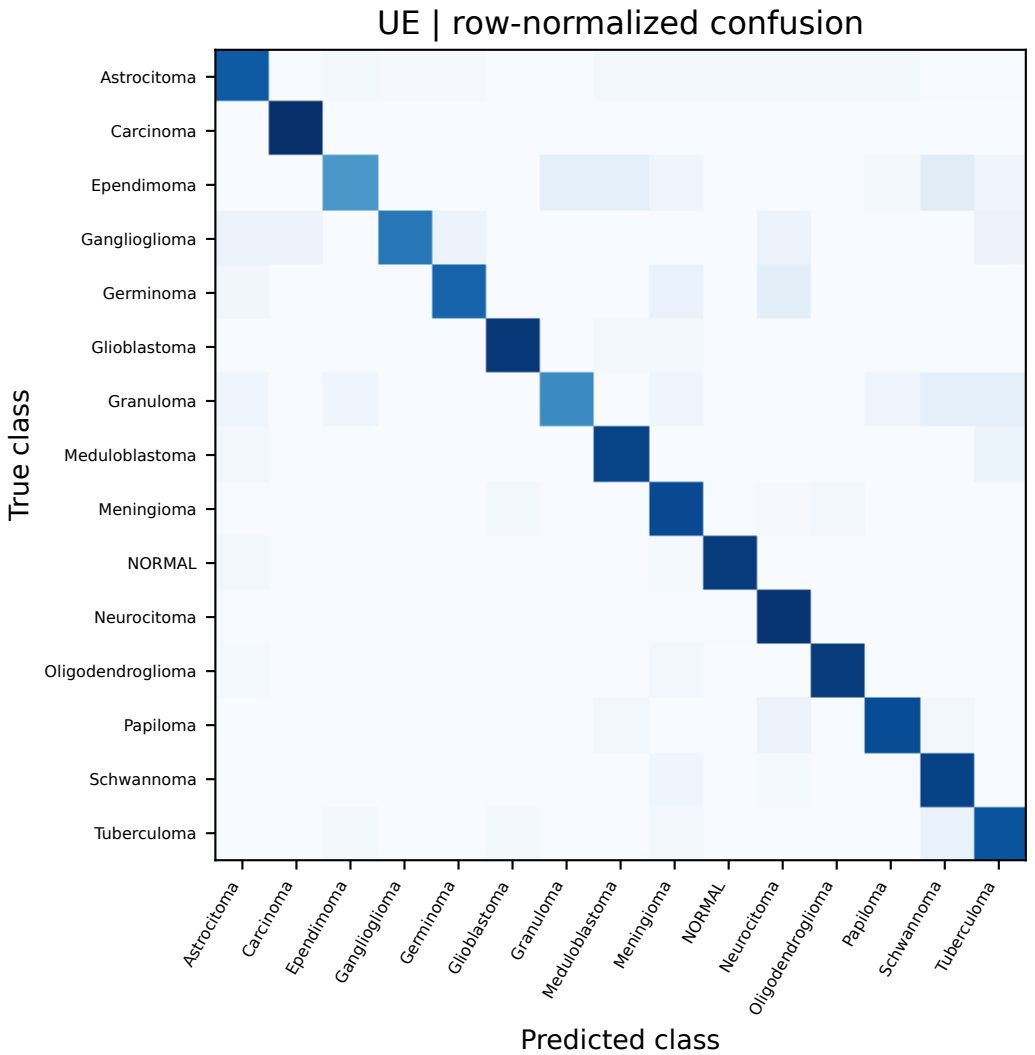


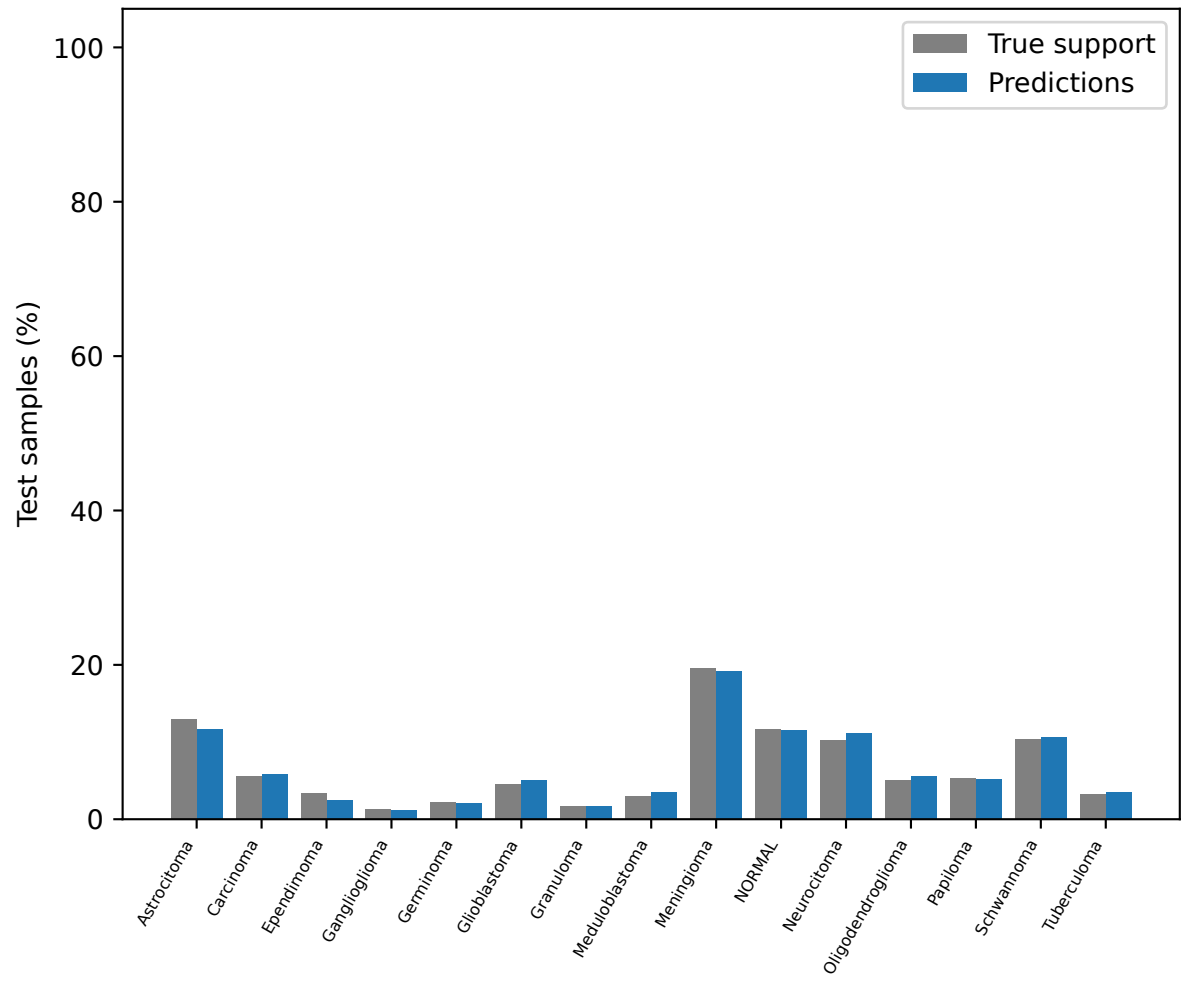


MRI15 | LSPAUG

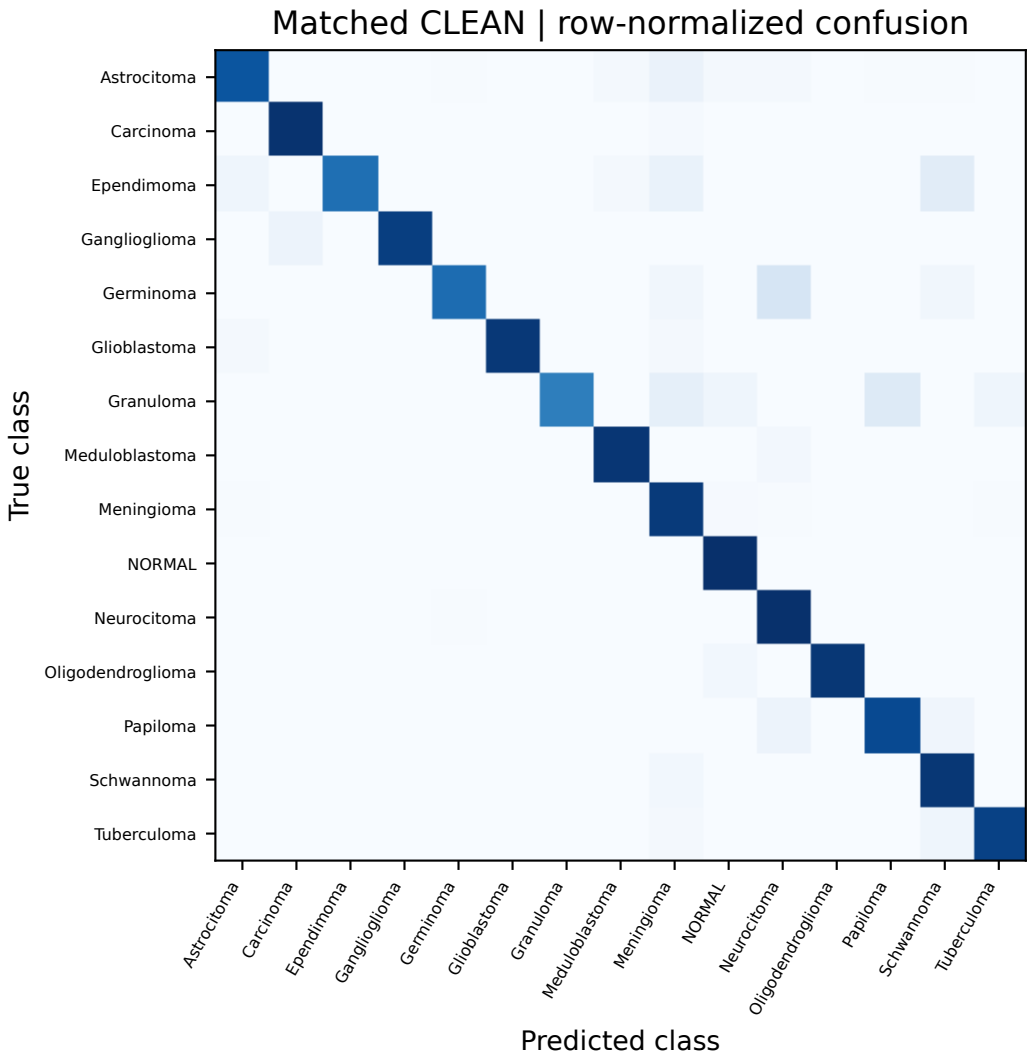


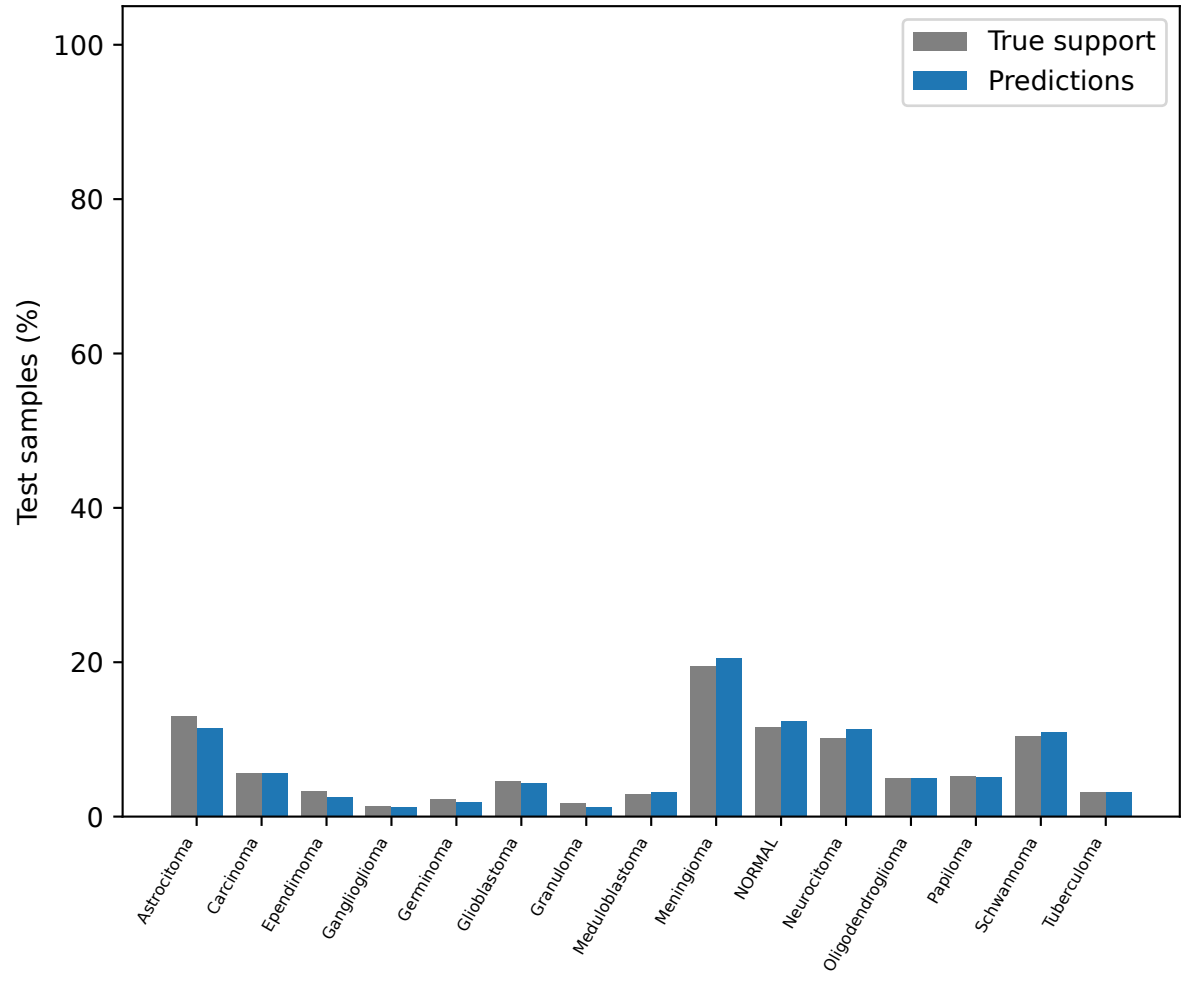


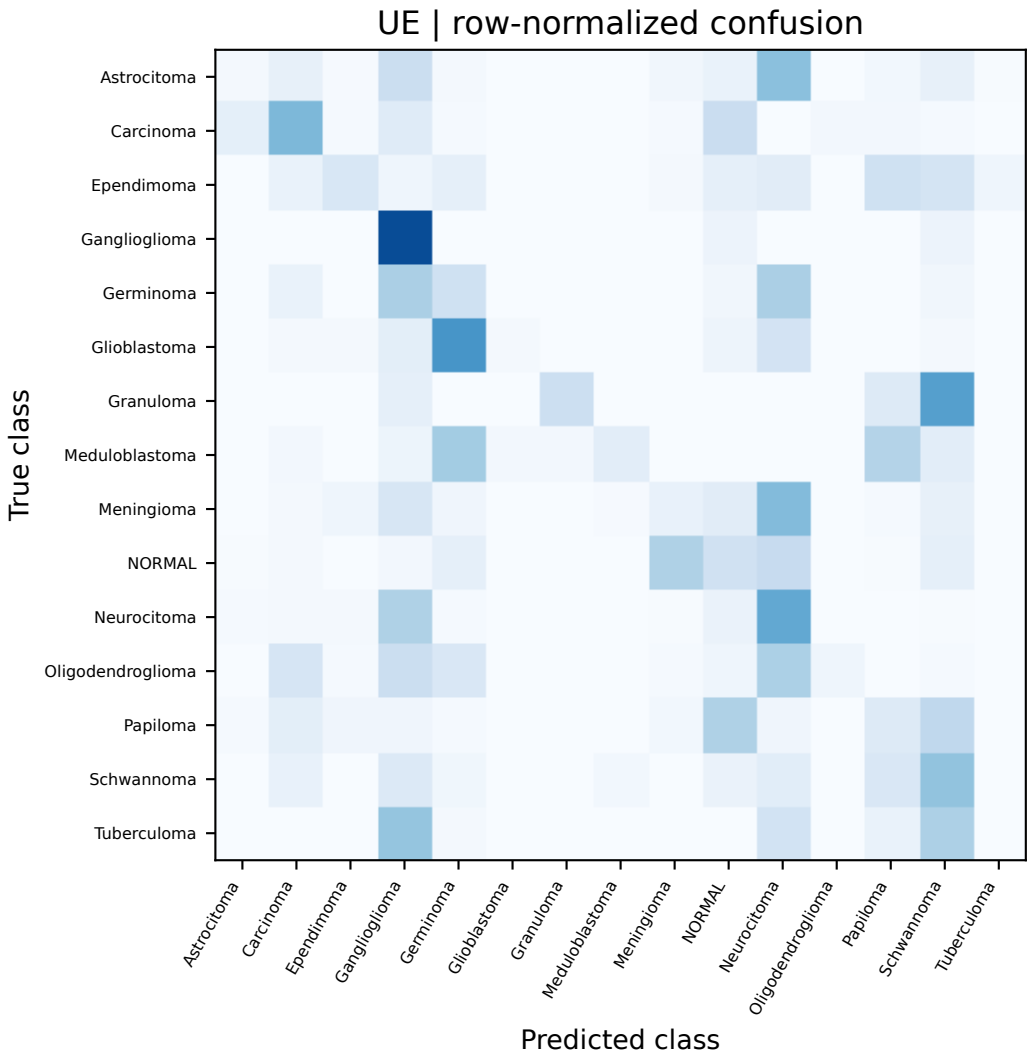


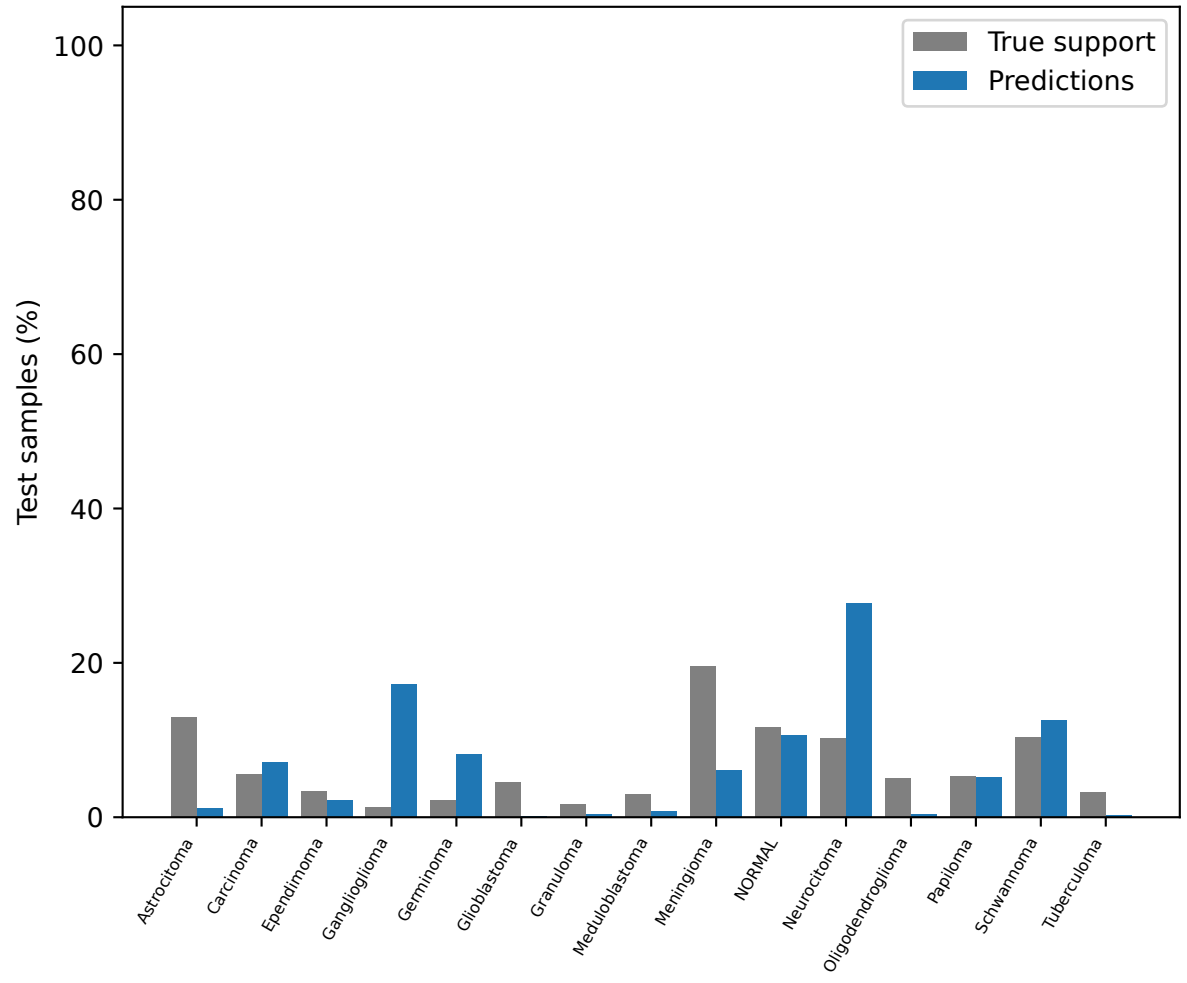


MRI15 | T-AP

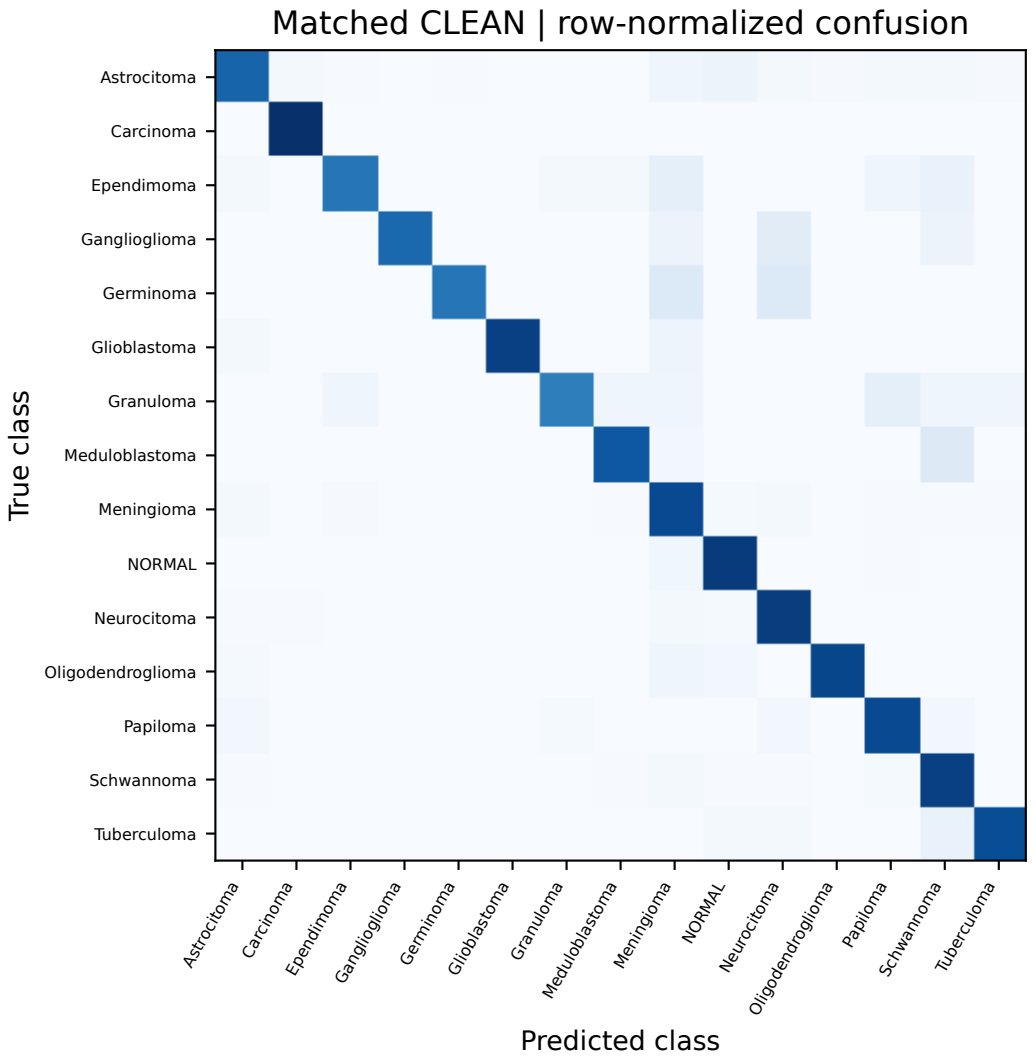

Matched CLEAN | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

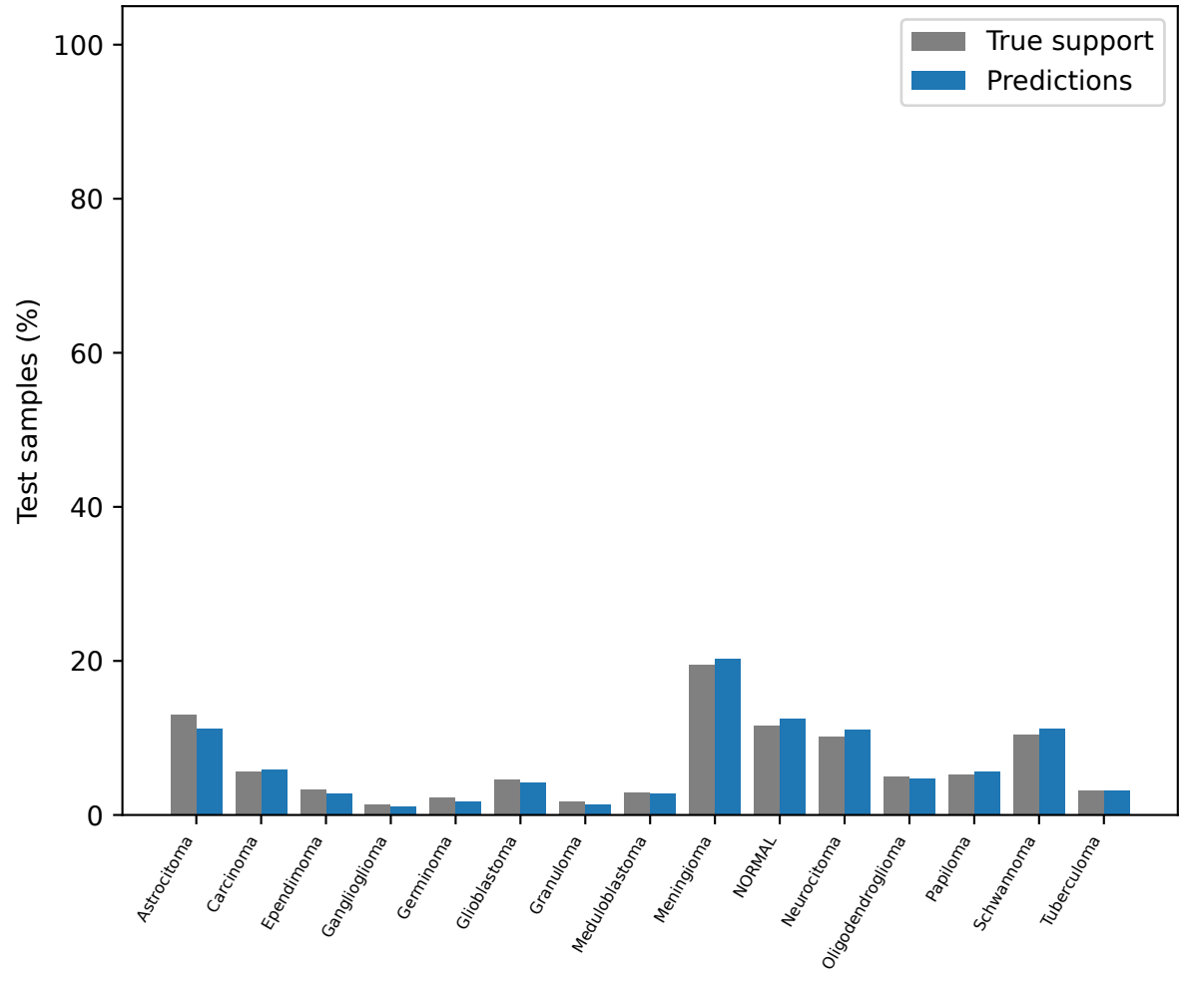

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

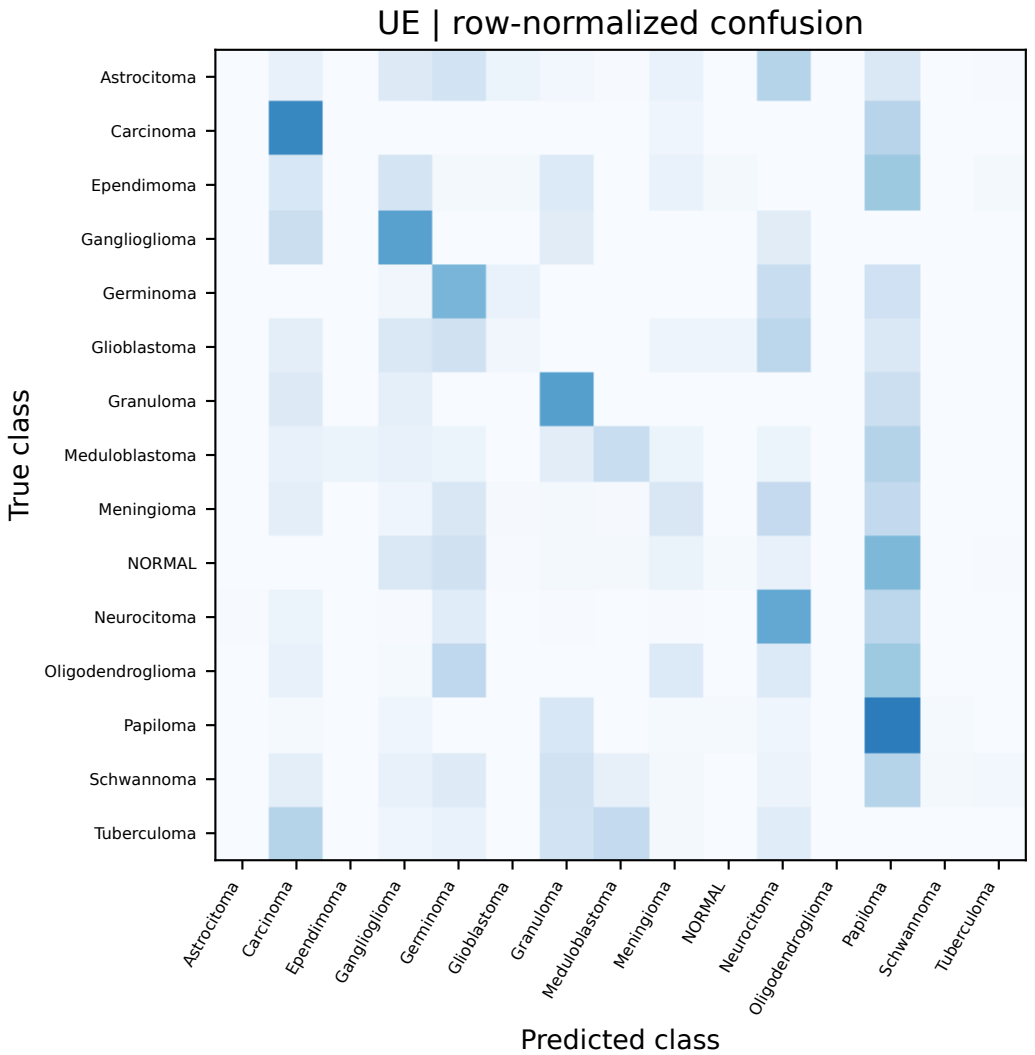

UE | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

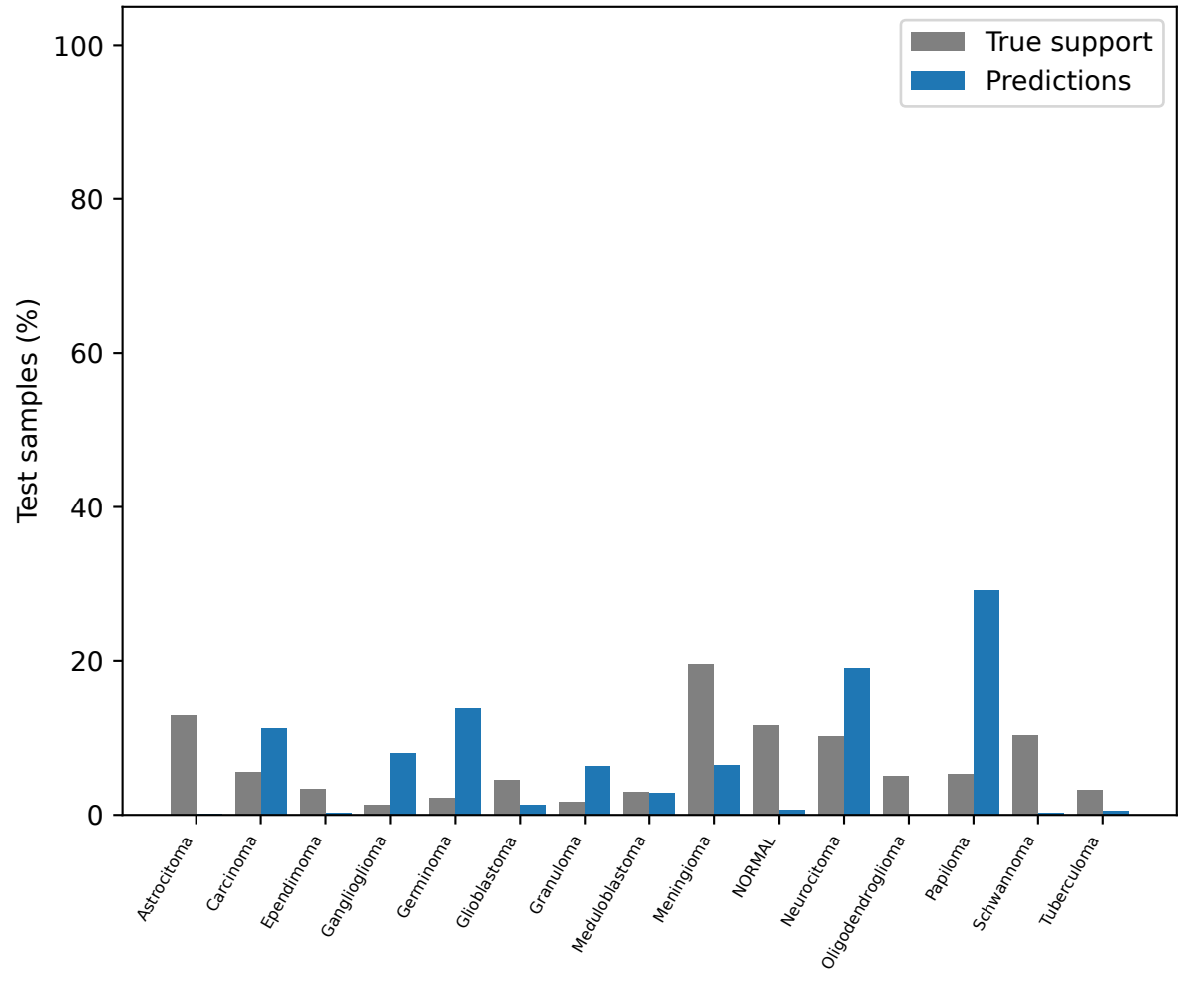

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

MRI15 | UT-AP

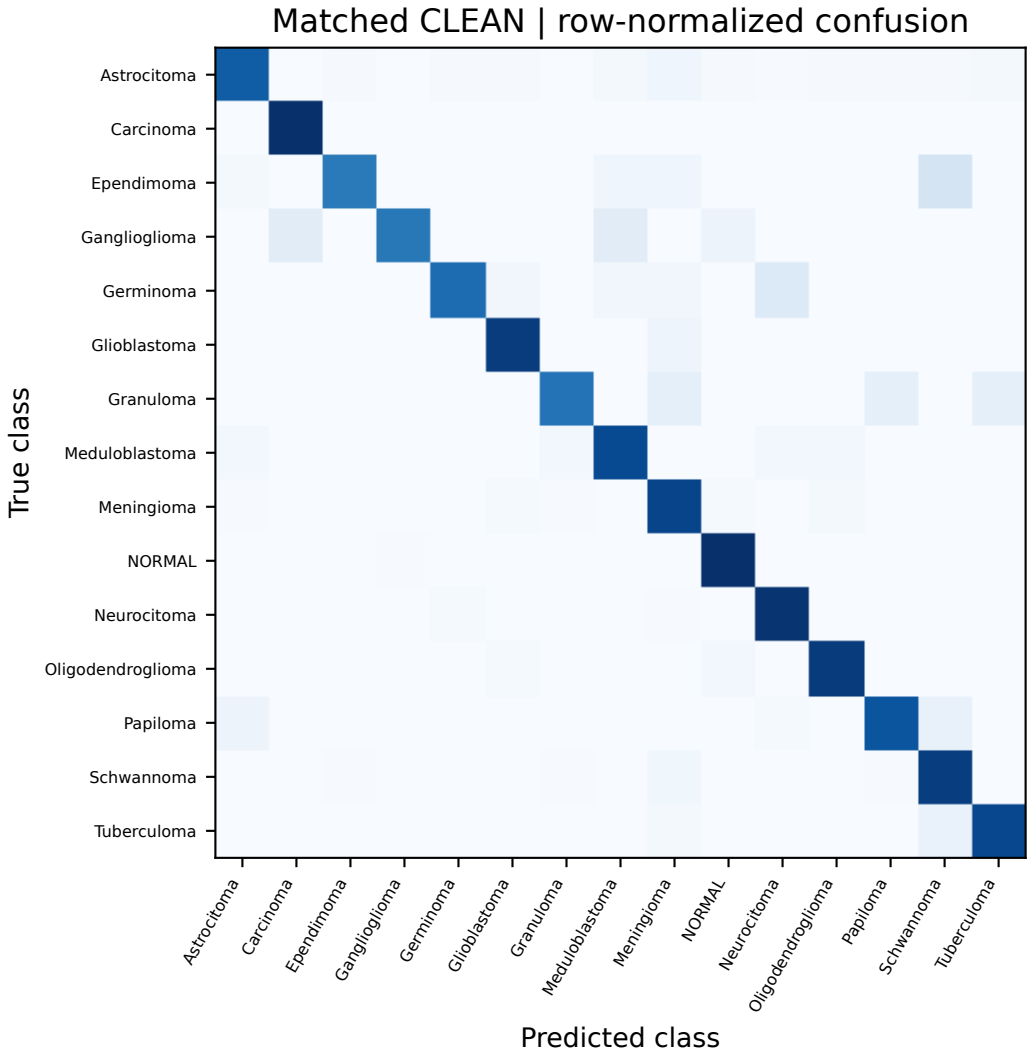

Matched CLEAN | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

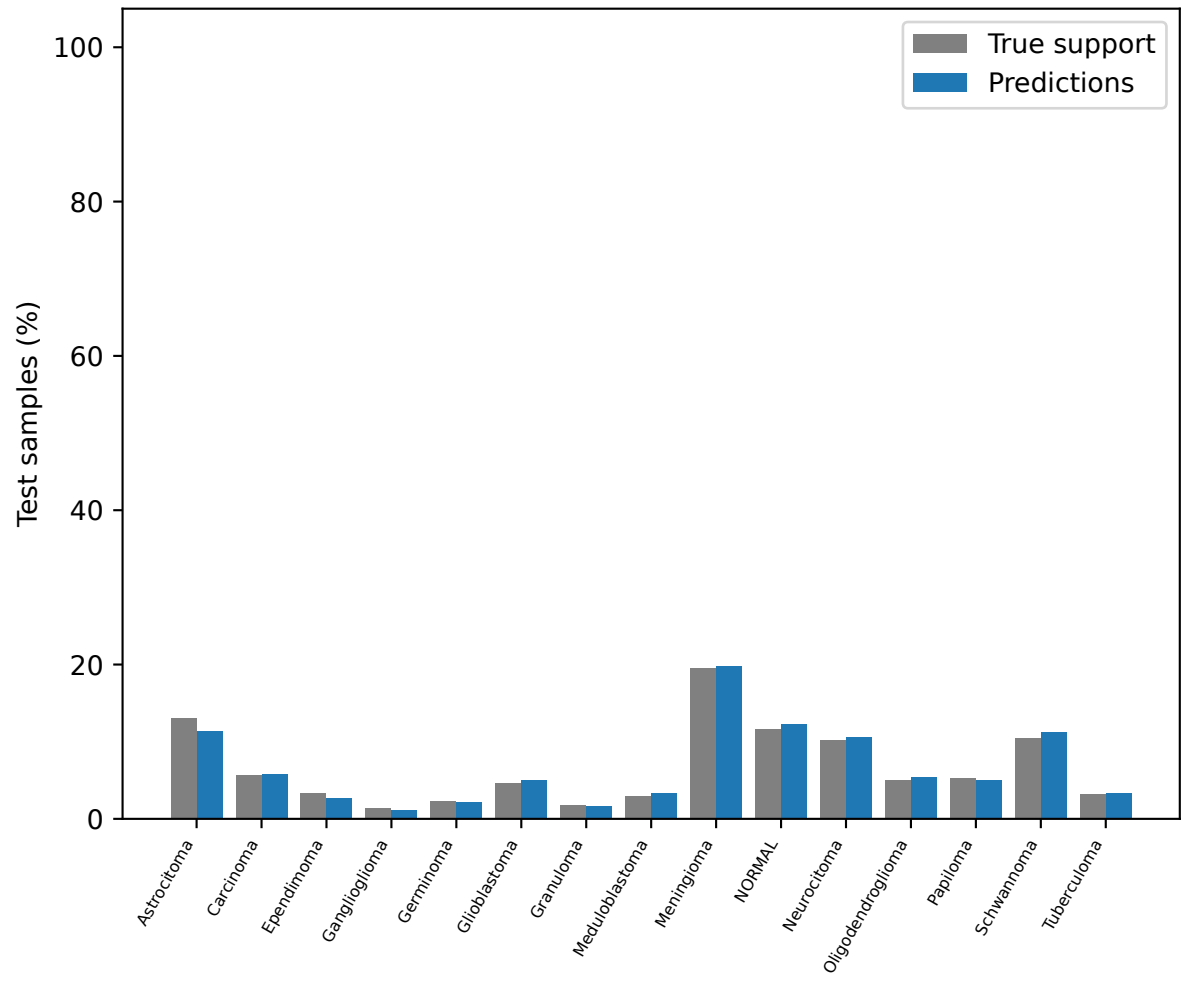

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

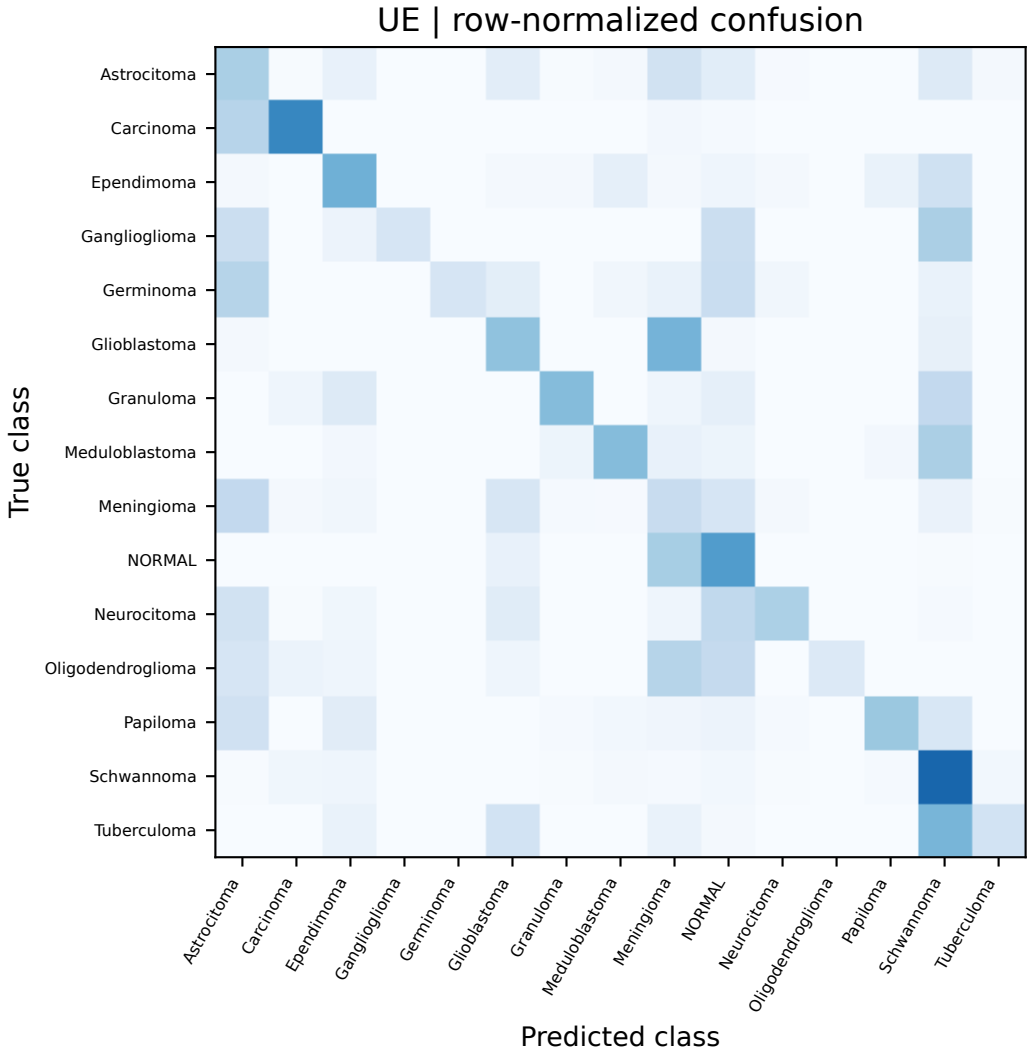

UE | row-normalized confusion
True class
Predicted class
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

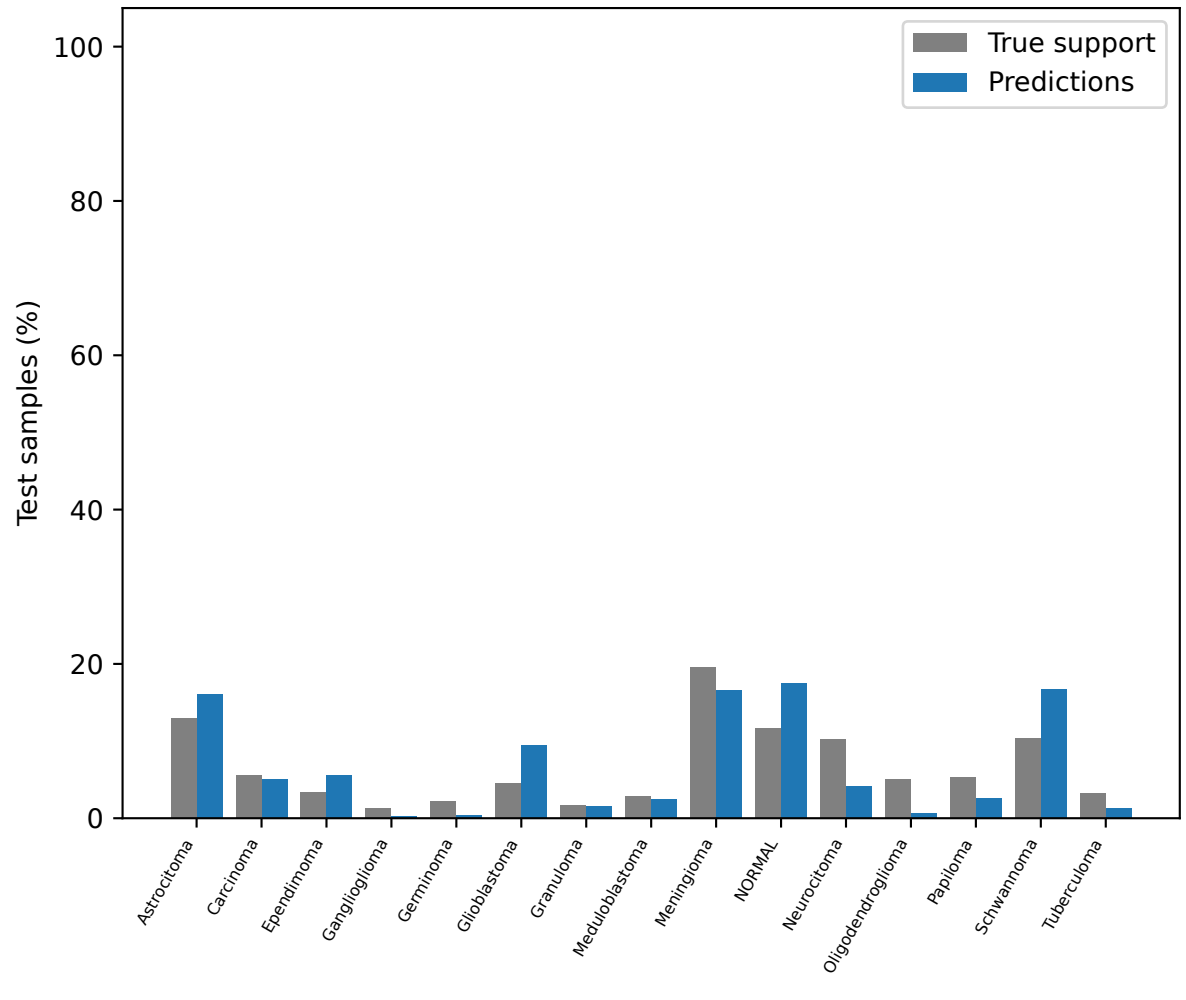

True support
Predictions
Test samples (%)
0
20
40
60
80
100
Astrocitoma
Carcinoma
Ependimoma
Ganglioglioma
Germinoma
Glioblastoma
Granuloma
Meduloblastoma
Meningioma
NORMAL
Neurocitoma
Oligodendroglioma
Papiloma
Schwannoma
Tuberculoma

MRI15 | HYPO_P1

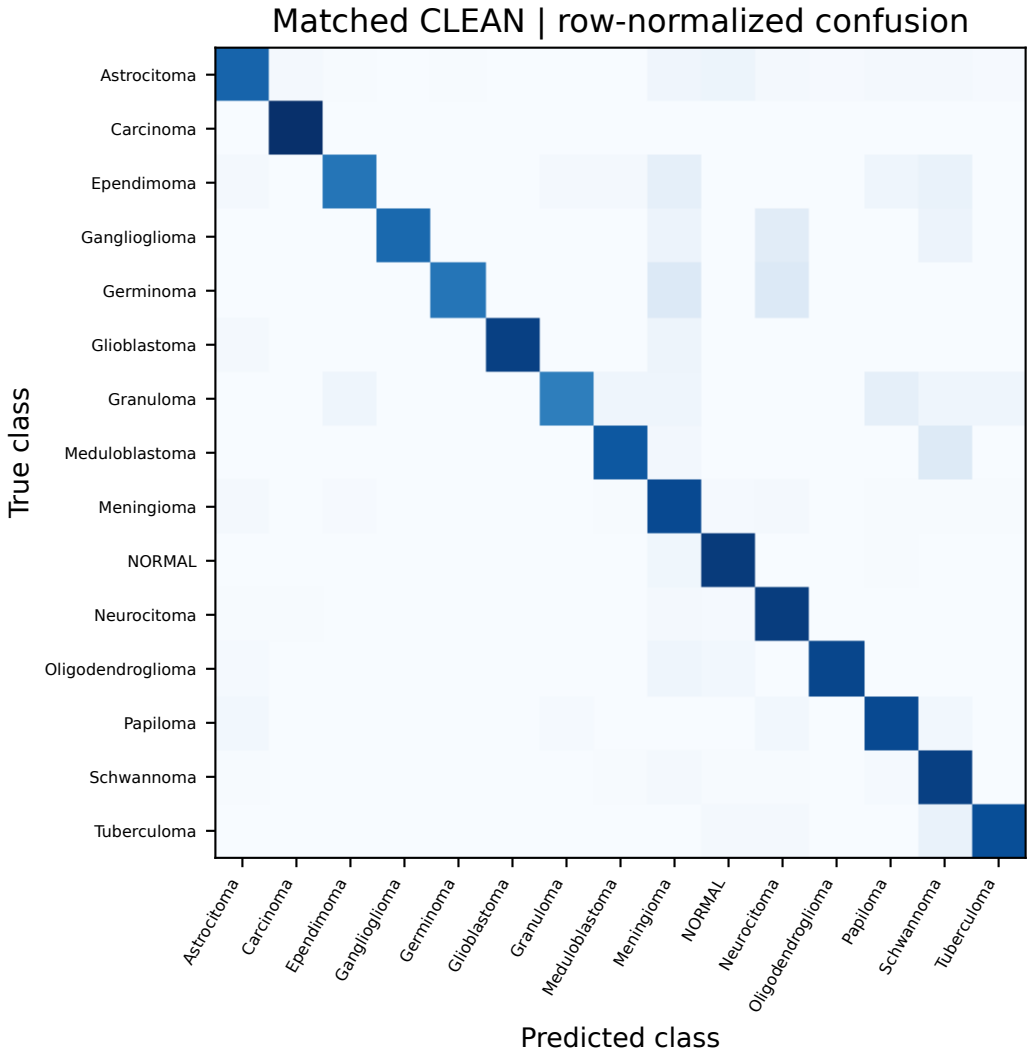


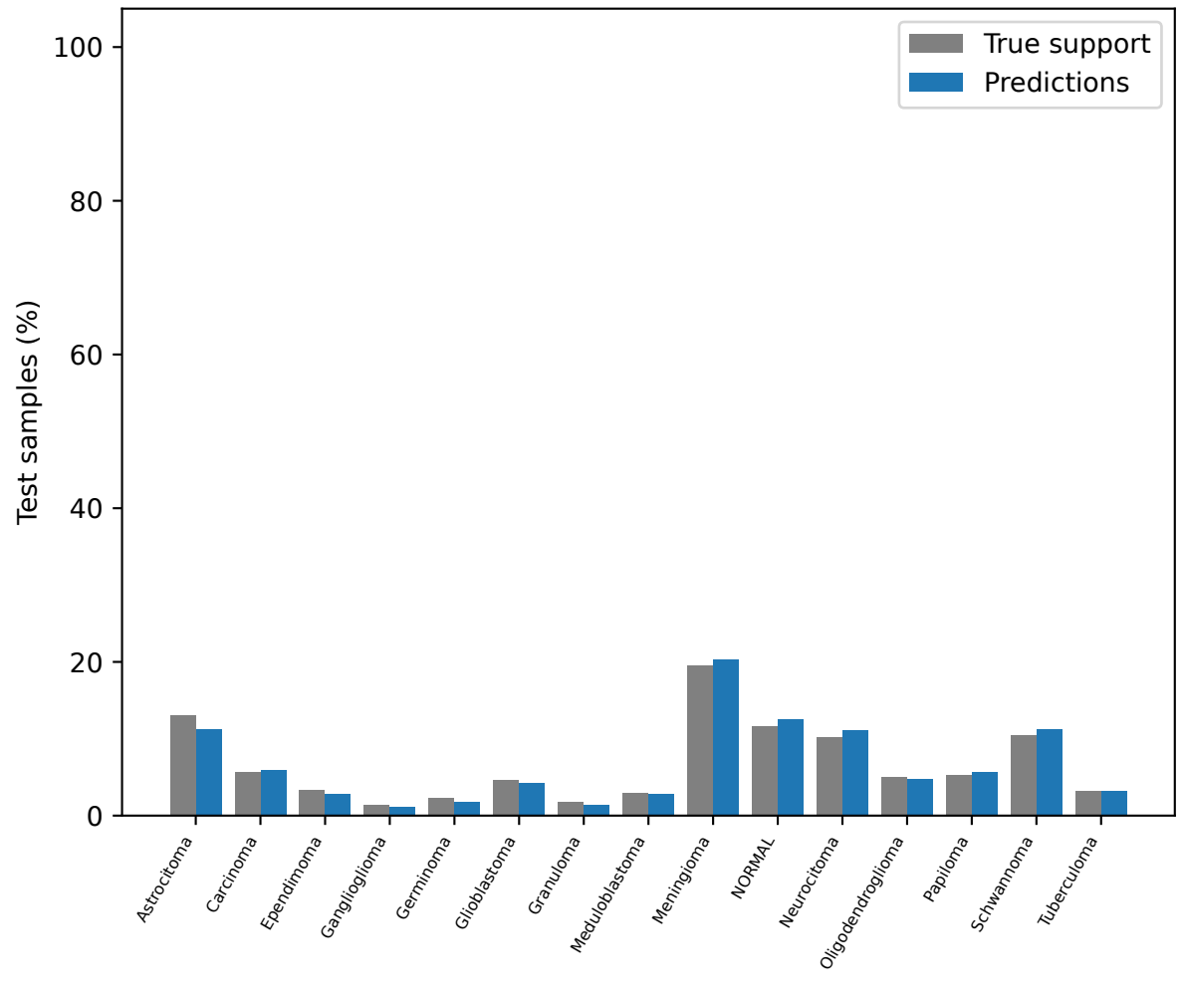


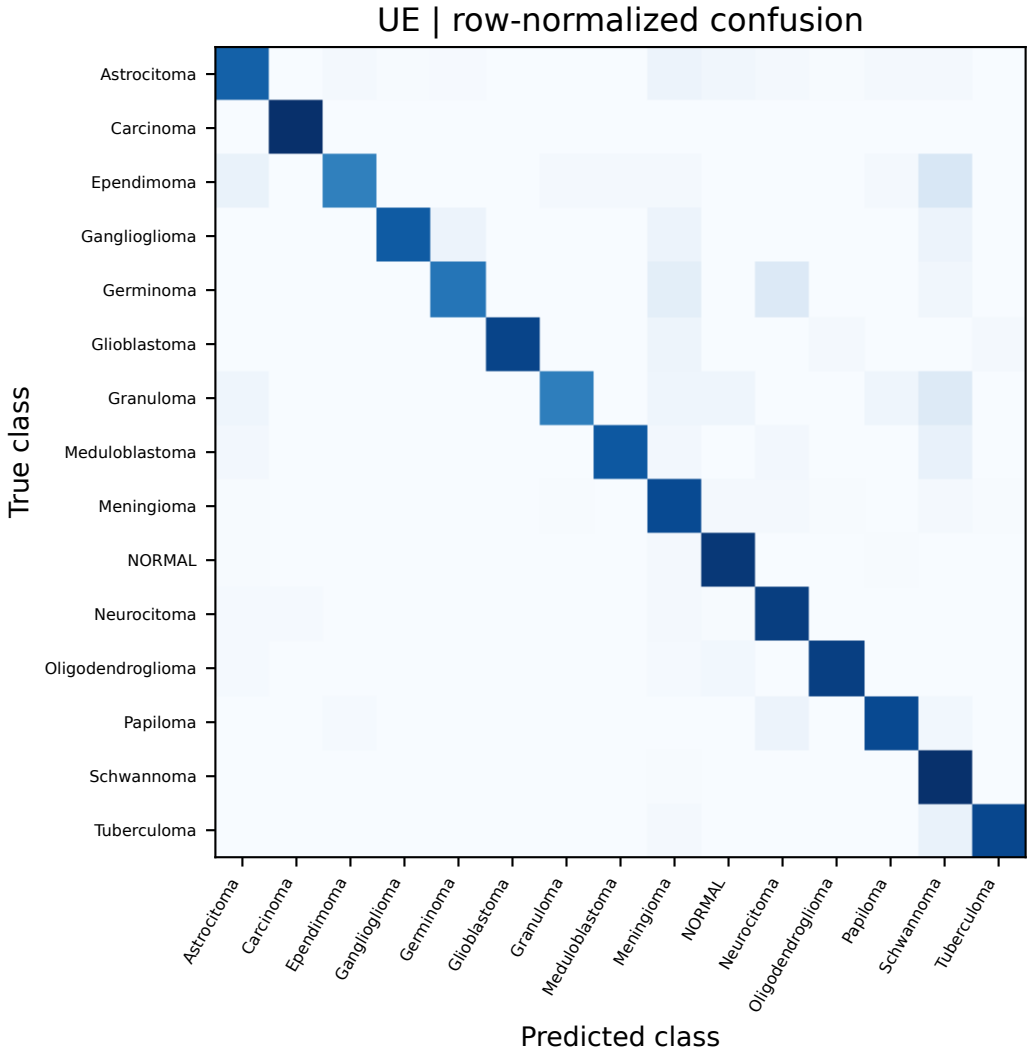


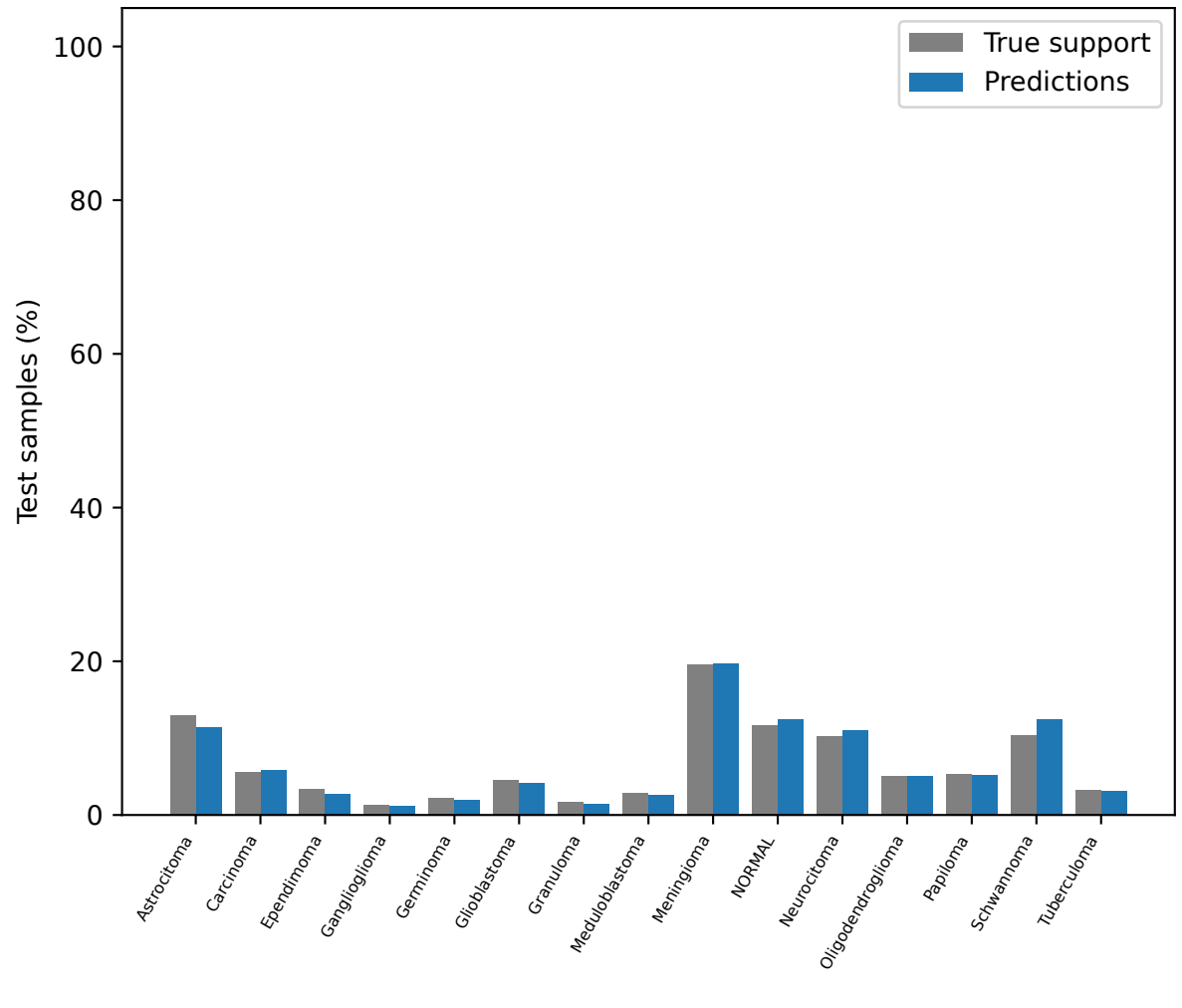


MRI15 | HYPO_P3









MRI15 | HYPO_P5

